

Designing Net-Zero Manufacturing Ecosystems by Integrating Industry 5.0 and Circular Product Lifecycle Management: A Systematic and a Bibliometric Review

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Abstract

Manufacturing industries are under mounting pressure to achieve net-zero emissions while maintaining competitiveness. The drive toward net-zero carbon emissions in manufacturing demands a cohesive strategy that combines circular economy principles with cutting edge industrial innovation. Therefore, this paper presents a multilayer integrated strategic framework for net-zero manufacturing ecosystems by integrating closed loop circular-economy strategies with the emerging Industry 5.0 (I5.0) paradigm and Circular Product Lifecycle Management (CPLM). We used Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and bibliometric approach to integrate I5.0's human-centric and sustainable orientation, and Product Lifecycle Management (PLM) practices identified across eight value chains, five support and net zero categories. The results demonstrate that the potential of I5.0 enabling technologies and human-centric innovation enhance circular lifecycle practices, encompassing design and sourcing through production, use, and end-of-life recovery to minimize waste and emissions. Whereas the bibliometric results reveal the dominant thematic clusters, keywords co-occurrence, influential publication outlets, and top contributing authors net-zero manufacturing ecosystems, I5.0 and CPLM scholarship over the past decade. Further, the proposed framework is structured into value chain stages aligned with the 10R circular strategies, underpinned by organizational and technological enablers that collectively promote a climate-neutral, resource-efficient manufacturing ecosystem. Moreover, we outline the categories of practices, their alignment with circularity loops and digital technologies, and discuss the synergistic outcomes of this integration including improved resource efficiency, reduced carbon footprint, and enhanced competitiveness. This study offers implications for manufacturing strategists to adopt human-centric, technology-enabled circular practices, and outlines supportive policy measures to foster collaborative, net-zero industrial ecosystems. This study contributes to existing body of knowledge by proposing a multidimensional net-zero manufacturing ecosystem which offers new insights by integrating diverse concepts into a comprehensive model that aligns with manufacturing technology management and sustainability objectives. Moreover, it provides a strategic roadmap for practitioners engaged in sustainable industrial transformation.

Keywords: Net-zero; circular economy; sustainable manufacturing; Industry 5.0; resources retention; circular product lifecycle management

1. Introduction

Manufacturing industries worldwide face unprecedented pressure to reduce their environmental impact and achieve net-zero carbon emissions in line with global climate targets. This challenge has spurred interest in the Circular Economy (CE) as a pathway to decouple industrial growth from resource consumption and waste generation (Ikram, 2026). CE strategies ranging from designing out waste to recycling end-of-life materials hold promise for addressing the 45% of greenhouse gas emissions tied to products and materials that clean energy measures alone cannot eliminate (Macarthur, 2019). Simultaneously, advanced manufacturing is undergoing a digital transformation, evolving from the Industry 4.0 (I4.0) era of automation and data exchange into Industry 5.0 (I5.0), a paradigm that places greater emphasis on sustainability, resilience, and human-centric innovation (Rahman *et al.*, 2025). However, there is a notable gap in both theory and practice: integration of CE and net-zero principles with the technological and human-centered advancements of I5.0 remains underdeveloped. To date, sustainability initiatives in manufacturing often proceed on separate tracks one focusing on circular lifecycle management of products, and another on adopting digital technologies for efficiency without a unified strategic framework to merge these efforts toward the overarching goal of net-zero emissions.

The need for integration is stressed by several developments. First, policymakers are explicitly linking digital innovation with green objectives. The European Union's I5.0 vision calls for leveraging research and innovation at the service of the transition to a sustainable, human-centric and resilient industry (European Commission, 2021). This recognizes that I4.0's technology-driven focus may be insufficient on its own for advancing sustainable development objectives, while I5.0 emphasizes production systems that are more attentive to planetary boundaries and worker well-being. The recent policy frameworks such as the EU's Net Zero Industry Act set ambitious targets for domestic clean technology manufacturing, explicitly aiming to redesign industrial value chains for climate neutrality by 2030 (European Commission, 2023; Mayer *et al.*, 2024). Achieving these targets requires not only decarbonizing energy sources but also fundamentally rethinking product lifecycles and resource flows in manufacturing. Moreover, manufacturing firms are beginning to explore how digital transformation can facilitate sustainability by using data analytics, Internet of Things (IoT) sensors, and AI to optimize energy and resource use but often lack guidance on aligning these tools with CE practices and net-zero goals (Peretz-Andersson *et al.*, 2024; Presciuttini *et al.*, 2024; Narkhede *et al.*, 2024). The literature reflects this fragmentation: studies on digital manufacturing focus mainly on I4.0 technologies with relatively little attention to the newer I5.0 concept, while research on circular manufacturing emphasizes 3R or 9R strategies but seldom considers how advanced industrial technologies and human machine collaboration can accelerate circular transitions. This siloed approach leaves a theoretical and practical gap in conceptualizing manufacturing ecosystems that are both digitally transformed and circular by design.

1.1 Linkage of Circular Economy with R0-R9 Framework

The concept of the CE has gained prominence as a model for sustainable industrial development, offering an alternative to the linear take-make-dispose economy (Dinda, 2020). A CE seeks to keep products, components, and materials in use for as long as possible through strategies that eliminate waste and continually reuse resources. In practical terms, the CE is often operationalized through a series of R-strategies or resource value retention options (ROs) that companies and supply chains can implement to increase circularity (Muller *et al.*, 2024). These

strategies are commonly enumerated as a hierarchy from R0 to R9, ordered from the most preferred preventing waste at source to least preferred recovering energy from residual waste. Table 1 summarizes the ten R-strategies in this loop-based framework, along with their core principles. The higher-level R's (R0-R2) represent short loops focused on avoiding resource use in the first place, the mid-level R's (R3-R7) are medium loops that extend product lifespans, and the lower R's (R8-R9) are long loops dealing with material recovery at end-of-life.

Table 1. The 10R framework of circular economy strategies (R0-R9)¹

R# (Strategy)	Description	Practice
R0-Refuse	Avoid use of materials or products altogether by eliminating unnecessary consumption or substituting services for products. This may involve dematerialization or offering alternative solutions so that a product is not needed.	Designing product-out service's digital music streaming instead of physical CDs, or simply not producing low-value, wasteful products.
R1-Rethink	Rethink the product use or design to make it more efficient or multifunctional, and to question whether the same function can be met with less material or through sharing and servicing models. Often involves intensifying product use shared use, and product-as-a-service.	Car-sharing programs, higher utilization of vehicles or modular product design that allows one product to serve multiple purposes.
R2-Reduce	Reduce material and energy consumption in production and use. Achieve the same function with fewer resources by increasing efficiency or using lighter, more sustainable materials.	Lean manufacturing to minimize scrap, lightweight product design, and energy-efficient manufacturing processes.
R3-Reuse	Reuse products by using them again for the same purpose by either the same or a new user, with minimal processing. This keeps products at their highest utility without major refurbishment.	Second-hand sales or take-back and re-sale of used equipment refurbished electronics, sold as used products.

¹ Note: R0 = Refuse; R1 = Rethink; R2 = Reduce; R3 = Reuse; R4 = Repair; R5 = Refurbish; R6 = Remanufacture; R7 = Repurpose; R8 = Recycle; R9 = Recover

R4-Repair	Repair and maintenance to fix a faulty or broken product so it can be used longer. This extends the product's life with minor restoration, delaying disposal.	Repair services for appliances, offering spare parts and repair manuals to customers to extend product longevity.
R5-Refurbish	Refurbish involves restoring an old product to good working condition by replacing or repairing major components. The product is updated, and its lifespan is significantly extended, though it remains in its original form.	Factory refurbishment of returned equipment, such as certified refurbished smartphones, computers, or machinery restored to good working condition for resale
R6-Remanufacture	Remanufacture means using a product's parts in a new product of the same type. It often entails disassembling a product, cleaning and repairing components, and reassembling them to as-new specifications.	Automotive engine remanufacturing where usable parts from end-of-life engines are recovered and built into "good as new" engines.
R7-Repurpose	Repurpose is using a discarded product or its components for a purpose different from its original one, without extensive processing. It gives a new function to old items.	Using retired electric car batteries for stationary energy storage or converting shipping containers into construction modules.
R8-Recycle	Recycle materials to obtain raw material inputs for new products. This involves processing waste into materials through melting, shredding that can substitute virgin resources.	Plastic or metal recycling: collected products are processed to recover raw plastic pellets or metal alloys for manufacturing new items.
R9-Recover	Recover energy or useful by-products from waste that cannot be recycled. Often implies incineration with energy recovery or capturing value like biogas from organic waste.	Waste-to-energy incineration generating electricity, or anaerobic digestion of food waste to produce biogas fuel.

1.2 I5.0: Human-Centric and Sustainable Manufacturing

I5.0 is a relatively new concept that builds upon the digital foundations of I4.0, explicitly incorporating goals of sustainability, human-centricity, and resilience in industrial systems. Whereas I4.0 is characterized by the automation and data exchange technologies such as IoT, Artificial Intelligence (AI), robotics, and big data analytics aimed at creating smart factories (Soori *et al.*, 2023). I5.0 is often described as a paradigm shift that places research and innovation at the

service of people and the planet. It encourages manufacturing technologies to be developed and applied in ways that are more closely aligned with environmental sustainability and social well-being, while extending the focus beyond operational efficiency and productivity gains (Xu *et al.*, 2021). This vision emerged from recognition that I4.0 alone may not adequately address global challenges like climate change and inequality; as noted by the European Commission in launching I5.0, achieving the Sustainable Development Goals (SDGs) requires expanding the scope of industrial transformation to explicitly include people-planet-prosperity considerations (Czvetkó *et al.*, 2025).

The key characteristics of I5.0 can be distilled into three interrelated pillars: human-centricity, sustainability, and resilience. Human-centricity means that technology should augment and empower human workers rather than replace or subordinate them. I5.0 envisions close collaboration between humans and machines such as operators working alongside collaborative robots (cobots), using exoskeletons to reduce fatigue, or leveraging AI decision-support systems to create a more efficient and creative production process that values human input (Ivanov, 2023). This approach acknowledges that humans bring creativity, adaptability, and ethical judgment, which, when combined with intelligent automation, can lead to better outcomes than automation alone. Moreover, it highlights worker well-being; in an I5.0 setting, the work environment and tasks are designed to improve safety, ergonomics, and job satisfaction (Battini *et al.*, 2022).

Sustainability in I5.0 extends the digitalization agenda to explicitly target environmental objectives. I5.0 companies are expected to adopt technologies and practices that reduce ecological footprint from cutting energy usage and emissions to minimizing waste and pollution in production. Importantly, this pillar links to the CE (Dhayal *et al.*, 2025). I5.0 stresses producing exactly what is needed, when needed, thereby avoiding overproduction and excess inventory that might later become waste. Advanced forecasting, mass customization, and flexible production enabled by I4.0 technologies can help align production more closely with actual demand, thereby reducing surplus output and unnecessary resource consumption (Grabowska *et al.*, 2022).

The third pillar, resilience, refers to strengthening the ability of manufacturing systems to withstand and adapt to disruptions, whether they are economic, environmental, or social. I5.0 promotes resilience through diversification of supply chains, flexible production networks, and the use of technologies to anticipate and respond to potential disruptions such as AI for predictive maintenance or blockchain for supply chain transparency and risk management (Ivanov, 2023). This is highly relevant for sustainable manufacturing because climate change and resource scarcity are sources of disruption that resilient strategies must address. In essence, the resilience aspect provides another synergy with the CE: closed-loop and regionalized resource flows tend to be more resilient by design, as they rely less on continuous input of scarce resources (Gargari *et al.*, 2025). In comparing I4.0 and I5.0, a few differences stand out that are pertinent to our framework. I4.0 centers on automation, efficiency, and connectivity it gave us smart machines, factory automation, and vertically and horizontally integrated IT systems for production. Table 2 provides a brief comparison of the two paradigms, highlighting aspects relevant to sustainability and circularity.

Table 2. I4.0 versus I5.0 Key distinctions in focus and objectives

Aspect	I4.0 (Technology-Centric)	I5.0 orientation of digital and organizational capabilities
Primary Focus	Automation, efficiency, productivity through advanced technology integration (smart factories).	Human-centric innovation, sustainability (environmental and social), resilience of systems.
Role of Humans	Humans are users/operators of largely automated systems; emphasis on reducing human intervention (lights-out manufacturing).	Humans are positioned central collaborators with machines, with technology used to augment human skills and support worker well-being.
Technology Use	IoT, AI, robotics, and analytics to optimize manufacturing processes for cost and speed. Often technology deployment is ends-oriented (productivity as end goal).	Digital capabilities derived from the I4.0 trajectory, including AI, IoT, robotics, analytics, digital twins, and connected platforms, are implemented with specific human-centric, sustainability, and resilience goals in mind. Technology is regarded as a facilitating tool rather than the ultimate goal of industrial transformation.
Production Philosophy	Mass customization and intelligent automation to meet demand efficiently; risk of overproduction if demand forecasting errs.	Personalized production closely aligned to actual societal needs, avoiding excess. “Produce exactly what is needed” to reduce surpluses and waste. Emphasizes adaptive production to prevent overproduction.
Sustainability Approach	Implicit or secondary efficiency gains can reduce resource use, but environmental goals not primary drivers in original I4.0 concept.	Explicit and core objective – design production systems for low carbon footprint, resource circularity, and minimal waste from the outset.
Resilience and Society	Focus on supply chain integration and flexibility, but less on social implications. Resilience considered in terms of redundancy and agility in operations.	Emphasizes resilience including social dimension (just transition for workers) and environmental shocks. Industry is seen as part of society; corporate responsibility and community impact are key (e.g., aligning with SDGs).

A fundamental distinction is maintained throughout this study between digital technological capabilities and the normative principles of I5.0. Technologies including the IOTs, AI, advanced analytics, cloud computing, robotics, digital twins, cyber-physical systems, and blockchain have largely emerged from or gained prominence within the digitalization trajectory of I4.0. I5.0 does not necessitate the reclassification of these technologies as a distinct technological generation. It alters the purpose, governance, and performance criteria for the deployment of digital capabilities by prioritizing human-centricity, sustainability, and resilience (European Commission, 2021; Pan et al., 2026; Grosse et al., 2023). This study

considers these technologies as digital or I4.0-origin enabling capabilities that, when intentionally configured, can enhance worker well-being and agency, environmental performance, resource circularity, and system resilience in support of I5.0.

1.3 Circular Product Lifecycle Management(CPLM) and Digitalization

PLM refers to the coordinated management of all data, processes, and decisions throughout the life of a product, from initial conception and design, through manufacturing and use, to end-of-life disposal or recycling (Scott, 2011). Traditionally, PLM has been employed as an enterprise strategy, often supported by specialized software, to integrate different stages of product development and improve information flows among R&D, engineering, production, and service domains (Hines *et al.*, 2006; Tai, 2017). In the context of sustainability and CE, PLM takes on an expanded significance: it becomes the backbone for CPLM, where the goal is not only to manage the product from cradle to grave, but to enable a cradle-to-cradle or regenerative lifecycle. Essentially, circular PLM means embedding CE principles into each phase of the product's life and using digital tools to track and optimize circular outcomes (Rusch *et al.*, 2023).

A core idea in circular PLM is creating a digital thread for the product, where all relevant information about the product's materials, components, and history of usage and maintenance is collected and accessible throughout its life. This digital record is crucial for making informed decisions at various stages such as knowing what materials are in a product and their condition at end-of-life can determine whether it's feasible to refurbish or recycle it. Concepts like the Digital Product Passport (DPP) exemplify this approach. A DPP is essentially a digital twin of the product's key attributes and life events, intended to facilitate circularity by documenting a product's journey from start to end (Zhang and Seuring, 2024).

Circular PLM strategies are being developed to support what some call a holistic circular product lifecycle. Circular strategies improve the product lifecycle, identifying where in product development, production, or use certain R-strategies can be applied (Aguiar *et al.*, 2022). In this regard, decisions made during the beginning-of-Life, design phase may support easier product disassembly and enable circular strategies such as R4 Repair, R5 Refurbish, R6 Remanufacture, through the use of recyclable or recycled materials (R8) (Alamerew and Brissaud, 2019). During the middle-of-life use phase, a PLM system can monitor performance to optimize maintenance schedules (R4) and track when the product might be suitable for remanufacturing or upgrading (R5-R6) (Mestre and Cooper, 2017). In the end-of-life phase, PLM systems integrated with reverse logistics can help sort products into reuse, remanufacturing, or recycling streams based on their condition and composition. Moreover, PLM comprises into slowing, narrowing, and closing loops are three broad principles that can be mapped onto these lifecycle phases: slowing loops (extending life) aligns with R3-R7 activities, narrowing loops (using fewer resources per product) ties to R0-R2 at design and process optimization, and closing loops (material recovery) relates to R8-R9 at end-of-life (Bocken *et al.*, 2016). A well-configured PLM system underpins all three by providing the data and process management needed to implement each strategy systematically.

1.4 Net-Zero Manufacturing Principles

The term net-zero manufacturing refers to the strategic aim of eliminating or fully offsetting all greenhouse gas (GHG) emissions from manufacturing operations and associated

value chains. Achieving net-zero means that any residual emissions which are technically or economically infeasible to eliminate are balanced by removal of an equivalent amount of CO₂ from the atmosphere, resulting in no net increase in atmospheric GHG levels (Ikram, 2026). This concept has gained prominence as countries and industries set mid-century climate targets such as, many nations have committed to net-zero emissions by 2050 under the Paris Agreement framework, and sectors like automotive, cement, and steel have outlined net-zero roadmaps. In manufacturing, net-zero principles involve a combination of decarbonization strategies: switching to renewable energy sources to cut scope 2 emissions, electrifying or otherwise decarbonizing industrial processes to reduce scope 1 emissions, improving energy and material efficiency reducing emissions per unit of output, and addressing supply chain and end-of-life emissions (scope 3) which often requires CE approaches (Vieira *et al.*, 2021).

A key principle in net-zero manufacturing is the integration of CE practices as a means to tackle the substantial portion of emissions embedded in materials and products. Research by material-focused organizations such as the Ellen MacArthur Foundation has highlighted that roughly 45% of global emissions come from the production of goods materials, products, food, whereas 55% come from energy generation and use (Macarthur, 2019). This implies that even a fully renewable energy system would only solve a little over half of the emissions challenge; the rest must be addressed through how we make and manage products. Thus, we can't achieve net zero without the CE, has become a mantra echoed in industry forums. Circular approaches like using recycled materials significantly cut emissions compared to using virgin materials since virgin material production such as steel from iron ore, or plastics from petroleum is typically very energy- and carbon-intensive (Santolini *et al.*, 2023). Similarly, prolonging the life of products means fewer new products need to be manufactured, directly avoiding the emissions that would have resulted from those new productions. A vivid demonstration is in electronics: refurbishing a computer avoids the emissions from manufacturing a new one, which can be dozens of kilograms of CO₂ for a laptop. Net-zero strategies thus explicitly encourage loops such as recycling and reuse to reduce embodied carbon in products. The net-zero perspective also values material efficiency getting the same output with less input, aligning with R2 Reduce, and substitution by using low-carbon or renewable materials (Tang *et al.*, 2023).

This study addresses the gap by developing a strategic framework for designing net-zero manufacturing ecosystems that integrates I5.0 and Circular Product Lifecycle Management (CPLM). Building on prior work by Ikram, (2026) on net-zero circular value chains (NZCVCM) presented in Figure 1, we adapt and extend those concepts to incorporate the I5.0 paradigm and an explicit loop-based circular structure (R0-R9). The framework is developed through a dual methodology: a systematic literature review Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) combined with a bibliometric analysis. PRISMA approach and proposes a integrated conceptual model wherein CE R-strategies are mapped across the manufacturing value chain and enabled by I5.0 technologies and practices toward achieving net-zero carbon outcomes. The proposed model integrates various domains, demonstrating how advanced digital tools, including IoT, AI, and robotics, alongside human-centric innovations such as co-creative design and worker augmentation, can strengthen circular strategies including reuse, remanufacturing, and recycling, thereby significantly minimizing waste and emissions within an

industrial ecosystem. This study expands on a category and practice outcomes that align specific value chain stages and support functions with the 10R circular strategies and I5.0 enablers. In doing so, we provide manufacturing strategists with a holistic view of how to simultaneously pursue digitalization and sustainability.

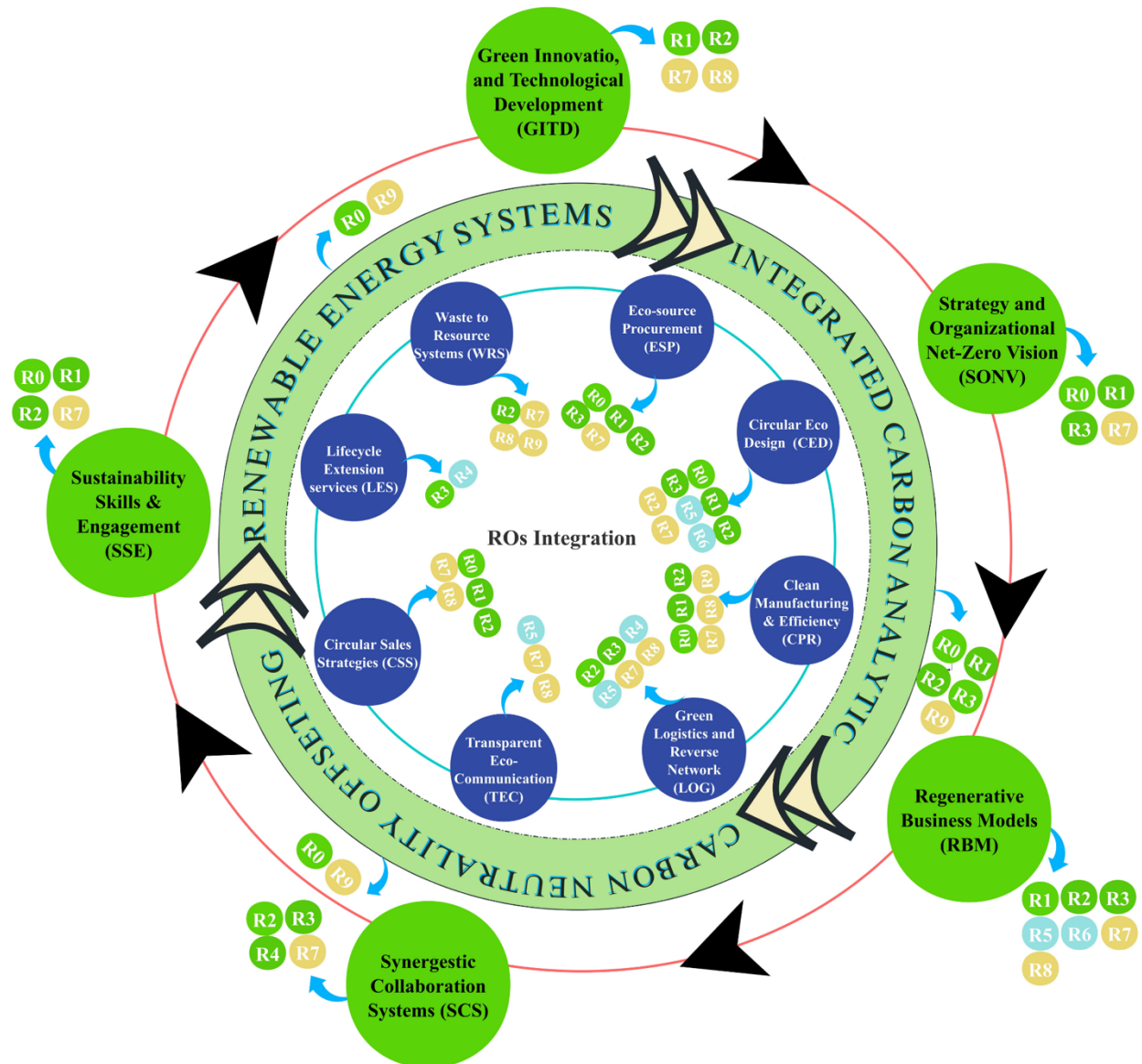


Figure 1: Net-zero circular value chain model (NZCVCM) model (Ikram, 2026)

The contributions of this study lies primarily in the extension and integration of established research streams rather than in the individual introduction of constituent concepts. First, it offers a novel conceptual integration of CE loops, I5.0 principles, and net-zero carbon objectives into one comprehensive framework, thereby advancing theoretical discourse on sustainable operations and I5.0. Second, it establishes a framework of well-defined categories and practices that managers can utilize as a strategic roadmap for implementation, linking with human-centric digitalization and circular lifecycle to specific manufacturing activities and decisions. third, this study examines the synergistic benefits of integration, including enhanced resource efficiency, potential for

innovation, and increased resilience, along with implications for stakeholders and directions for future research. Finally, the bibliometric approach allows us to consolidate the fragmented literature on net-zero manufacturing ecosystems integrating I5.0 and CPLM and examine the alignment with various Sustainable Development Goals (SDGs)

2. Material and Methods

The objective of this study is to develop a novel theoretical framework based on an integrative literature review and synthesis. The PRISMA methodology focuses on systematically collecting insights from various related fields, specifically CE with an emphasis on the R0-R9 framework and value chain applications, I4.0/I5.0 and digital manufacturing, PLM, and sustainability/net-zero strategy. These insights are then integrated to develop a cohesive model for net-zero circular manufacturing ecosystems. This approach is consistent with established methodologies in management research for theory development, utilizing existing literature for concept formation and the construction of new frameworks, including literature-based model building as presented by (Jaakkola, 2020). The process involved multiple stages:

2.1 Development of research questions

This study presents a strategic framework that integrates I5.0 with CPLM to facilitate net-zero manufacturing ecosystems. The aim is to conceptualize the application of advanced technologies and human-centric strategies in I5.0 throughout all phases of a product's lifecycle, from design and production to utilization and disposal, with the goal of minimizing waste and emissions. The framework integrates these perspectives to assist manufacturers and stakeholders in systematically redesigning their operations and value networks to achieve sustainability. Key research questions include:

RQ (1): What are the core components and enabling factors of a manufacturing ecosystem that achieves net-zero emissions?

RQ (2): How do I5.0 technologies and human-centric innovations facilitate CE practices throughout the product lifecycle?

RQ (3): What strategic actions should managers and policymakers take to implement and support such an ecosystem?

2.2 Locating studies and keywords strategy

We conducted extensive literature searches utilizing academic databases, including Scopus and Web of Science, alongside targeted keyword queries to identify relevant publications until the end of year 2025. The systematic literature review (SLR) which are known for broad coverage and are commonly used in I5.0, manufacturing eco system including product lifecycle, CE and net zero (Narkhede *et al.*, 2023; Kumar *et al.*, 2024; Narkhede *et al.*, 2025). Keywords included combinations of terms such as *CE, R framework or 9R/10R, I5.0, I4.0 and sustainability, digital transformation and circular, PLM AND circular, net-zero manufacturing, sustainable manufacturing strategy*. We also reviewed supplementary policy and industry sources, including

the European Commission's Industry 5.0 reports and the World Economic Forum's net-zero guidelines, to provide additional practical context. These sources complemented the academic literature but were not treated as part of the formal bibliometric or systematic review dataset. The sources were initially gathered, are conceptual papers, empirical studies case studies, surveys, and review articles. A total of 2,507 documents were initially identified as a result of this comprehensive search. The increased number indicates a wider scope that encompasses the emerging literature on I5.0/I4.0, product lifecycle, sustainable manufacturing strategies, as well as research on CE and net zero initiatives. Restrictions concerning language and document type were implemented for English-language journal articles to guarantee the quality and relevance of the sources. All retrieved references were imported into reference management software, and duplicate records from the two databases were eliminated prior to screening.

2.3 Selection /Evaluation criteria

The literature was evaluated for relevance and quality. We prioritized sources that directly addressed the intersections of our areas of interest, specifically studies on I4.0/5.0 and environmental sustainability, as well as CE implementations in manufacturing. Additionally, foundational papers were included that established key concepts, such as Ikram (2026) and Reike et al. on 10R, as well as the European Commission's I5.0 communiqué to support our definitions. Each source underwent analysis to identify key themes and findings. A thematic coding approach was utilized, categorizing sources into themes including "Circular strategies in manufacturing," "I5.0 definitions and pillars," "Digital technology enabling circularity," "Net-zero frameworks in industry," and "Integrated models and gaps." This thematic analysis identified patterns and consistent recommendations for the integration of digital technology with CE, as well as recognized research gaps concerning the underexplored role of I5.0 in sustainability (Garrido *et al.*, 2024). We observed the convergence and divergence of various literatures; both CE and net-zero literatures stress the importance of redesigning value chains, whereas digital manufacturing literature frequently neglects this broader sustainability context, indicating a gap in convergence.

The identification and selection of studies employed a multistage filtering process consisting of three consecutive screening rounds. The process systematically narrowed the initial collection of 2,507 records to a final set of studies that specifically addressed three research questions (RQs). Prior to substantive screening, duplicate records were identified thru bibliographic information and subsequently removed. All records were exported to R for the purposes of cleaning, normalization, and duplicate detection thru DOI matching, exact-title matching, and manual verification of near-duplicate variants. The reduction from 2507 to 1380 occurred during the preliminary filtering stage. Records were excluded if they were duplicates, dated outside the range of 2015-2025, not written in English, not classified as articles or reviews, or clearly outside the study scope based on the title. To enhance transparency, the final manuscript presents a category-level breakdown of the 1127 exclusions as follows: duplicates (n=196); non-article/review items (n=150); non-English records (n=132); and records outside the specified period or clearly irrelevant to the topic (n=649). This resulted in 1,380 unique records available for title and abstract screening. Articles were selected based on the presence of relevant combinations in their titles or abstracts that connect I4.0/5 with CPLM and sustainability concepts.

During the second screening stage, titles, abstracts, and author keywords were reviewed to assess their alignment with the study's focus on I5.0-enabled manufacturing ecosystems, circular product lifecycle practices, resource-retention strategies, digital technologies, and net-zero manufacturing objectives. At this stage, 553 records were excluded because they did not show a substantive connection with the core focus of the review. Excluded records mainly addressed general industrial digitalization, unrelated engineering optimization, broad sustainability policy, or technical applications without a clear link to lifecycle-based circular manufacturing, I5.0 principles, CPLM, or net-zero-oriented manufacturing practices. After the third filter, a further 587 records were excluded because they did not provide sufficient substantive or analytical engagement with CPLM, I4.0/I5.0, CE and net-zero within the scope of the review. This screening process resulted in 240 records being retained for eligibility assessment.

The final qualitative coding of the 240 included studies was supported by a calibration round to improve coding consistency. Both authors independently coded a subset of the sample using the main framework categories and associated practice-level dimensions. Inter-coder consistency for this calibration round was assessed using Cohen's kappa, yielding $\kappa = 0.89$, which indicates strong agreement between coders. After the codebook was refined, the remaining papers were coded and cross-checked. Any contested coding decisions were resolved through discussion and consensus.

The search strategy was designed to be broad during the identification stage in order to capture the interdisciplinary nature of net-zero manufacturing, I5.0, CPLM, and CE research. Studies were not required to address I5.0, CPLM, CE, and net-zero transition simultaneously at the initial retrieval stage. Instead, records were retrieved when they contained a clear manufacturing, lifecycle, digital transformation, or industrial-sustainability anchor together with at least one related sustainability domain, such as decarbonization, carbon emissions, net-zero manufacturing, CE, resource efficiency, PLM, sustainable manufacturing, or SDG-oriented industrial transformation. This broader retrieval logic explains why a substantial number of studies remained after the initial screening, even though a smaller group explicitly integrated circularity, lifecycle management, and net-zero framing within a single study.

Because the final review corpus included conceptual, empirical, and review-based studies, a light-touch quality appraisal was applied rather than a design-specific exclusion rule. The appraisal considered thematic relevance, methodological clarity, specificity of reported practices, and adequacy of supporting evidence. This assessment informed the interpretation and weighting of evidence in the synthesis, but studies were not excluded solely because they used different methodological designs. At each stage, specific inclusion and exclusion criteria were implemented, as summarized in Figure 2, which presents the PRISMA-based study-selection and bibliometric-analysis workflow.

To enhance methodological rigor and reduce selection bias, two reviewers conducted each screening round independently and later resolved discrepancies through discussion, reaching a consensus decision set. The initial screening involved evaluating titles and abstracts to select publications where I4.0/I5.0, or closely related digitalization concepts, were central and that explicitly addressed sustainable manufacturing themes, such as product lifecycle management and net-zero/zero-carbon initiatives within a sustainability context (Ghobakhloo *et al.*, 2025). Eligible

studies must focus on circular manufacturing systems and/or net-zero ambitions, demonstrating a substantial discussion of practices, interventions, or strategic approaches. Records were excluded when these themes were mentioned only incidentally, tangentially, or without clear relevance to the study's scope, where I4.0/I5.0 or net-zero served as background context rather than as a focal point of analysis. Subsequent to this step, 1,380 documents were retained.

The second filter entailed a focused analysis of the introduction and conclusion sections of the selected studies (Kumar *et al.*, 2024). The objective was to confirm that each paper presented at least one actionable item, practice, or implementable strategy connecting I4.0/I5.0 with the advancement of CE (CE) objectives and/or net-zero decarbonization goals within manufacturing settings. Studies were included if they provided specific practices, including implementation pathways, operational measures, case-based evidence, or managerial and technological strategies.

STEP 1

Development of Research Questions

- RQ1** What are the core components and enabling factors of a manufacturing ecosystem that achieves net-zero emissions?
- RQ2** How do Industry 5.0 technologies and human-centric innovations facilitate circular economy practices throughout the product lifecycle?
- RQ3** What strategic actions should managers and policymakers take to implement and support such an ecosystem?

STEP 2

Locating Studies & Keywords Strategy

Databased:
SCOPUS & WOS

Publication Year:
Dec 2014-December 2025

Inclusion Criteria:
The string must cover "title", "abstract",
"keywords" type "Article or Review" in
English.

Search string

TITLE-ABS-KEY (("Industry 5.0" W/2 "net-zero") OR ("Industry 5.0" W/2 "carbon neutral*") OR ("Industry 5.0" W/2 "sustainable manufacturing") OR ("Industry 4.0" W/2 "net-zero") OR ("Industry 4.0" W/2 "carbon neutral*") OR ("Industry 4.0" W/2 "circular economy") OR ("Industry 4.0" W/2 "sustainability") OR ("net-zero manufacturing") OR ("carbon neutral manufacturing") OR ("net-zero" W/2 "production systems") OR ("zero-carbon" W/2 "industrial systems") OR ("low-carbon" W/3 "smart manufacturing") OR ("circular economy" W/3 "Industry 5.0") OR ("circular economy" W/3 "Industry 4.0") OR ("digital twin*" W/2 "net-zero") OR ("cyber-physical" W/2 "carbon") OR ("human-centric" W/2 "sustainable manufacturing") OR ("Circular Product Lifecycle Management") OR ("CPLM" W/2 "net-zero") OR ("circular economy" W/2 "lifecycle management") OR ("circular manufacturing" W/3 "framework*") OR ("carbon neutral" W/3 "framework*") OR ("net-zero" W/2 "design*") OR ("circular economy" W/2 "carbon reduction") OR ("resource efficiency" W/2 "Industry 5.0") OR ("resource efficiency" W/2 "Industry 4.0") OR ("smart manufacturing" W/2 "carbon tracking") OR ("digital PLM" W/2 "decarbonization") OR ("decarbonized manufacturing") OR ("zero-emission" W/2 "factory*"))

Searches

The search was conducted in end of 2025.

Total articles

A total of 2507 documents were initially identified

Output

After first filter 1380 documents remained. The second filter excluded 553 documents. Finally, 240 articles met the criteria after third filter.

Mixed approach

Deductive and inductive approach was adopted. Nvivo 12 was used for coding qualitative data. Matlab was used for aggregating data

The outcomes of systematic literature review are reported in a organized to answer RQ1,RQ2 and RQ 3.

3 GOOD HEALTH AND WELL-BEING



4 QUALITY EDUCATION



7 AFFORDABLE AND CLEAN ENERGY



8 DECENT WORK AND ECONOMIC GROWTH



9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



10 REDUCED INEQUALITIES



11 SUSTAINABLE CITIES AND COMMUNITIES



12 RESPONSIBLE CONSUMPTION AND PRODUCTION



13 CLIMATE ACTION



17 PARTNERSHIPS FOR THE GOALS



Refuse



Reduce



Resell



Repair



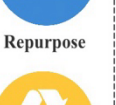
Refurbish



Remanufacture



Repurpose



Recycle



Recovery



Remine

STEP 3

Selection/ Evaluating Criteria

Filter 1

"Title & abstract"

Inclusion criteria

Studies core theme "Industry 4.0/5.0" and "circular product lifecycle"

Filter 2

"Introduction & conclusion"

Inclusion criteria

Studies that presents practice/strategy based on "Industry 4.0/5.0" and "net zero"

Filter 3

"Full text screening"

Inclusion criteria

Studies that presents practice based on "Industry 4.0/5.0" and "sustainable manufacturing" that a company/country can implement

STEP 4

Data Extraction Synthetic Analysis

Coding and categorization Industry 5.0 and circular vlaue chain practices

Codification of practices

CMPE: Clean manufacturing & process efficiency

WRR: Waster to resource cycling

Practice Categories

CMPE0301: Closed loop water cooling system align with R3

WRR1502: Advanced AI driven sorting technology

Coding and analyzing the relationship between Industry 5.0 practices, net-zero manufacturing

Examples

Keeping water, heat and energy circulating inside the plant. Start with flow mapping recovery loops

A smart recycling plant where scrap from manufacturing is sorted by alloy using AI-based sensors

CMPE0301 SDG7,12,13

WRR1502 SDG12,13,17

Structure

Table 3 presents circular value chain categories in net-zero manufacturing ecosystems with R-strategies and Industry 5.0 enablers

Table 4 describes Integrated framework support categories link organizational capabilities, Industry 5.0 concepts, and circular net-zero ambitions

Table 5 mapping the 50 integrated industry 5.0, net zero and sustainable manufacturing ecosystem align with circular RO and Sustainable Development Goals

STEP 5

Reporting Outputs

Figure 2: PRISMA-guided study-selection process and bibliometric-analysis workflow used in this study (authors own drawing)

This encompasses initiatives that promote reuse and recycling within supply networks, as well as the integration of renewable energy and efficiency measures targeting carbon neutrality. Studies were excluded if they were purely conceptual, addressed themes at a general level without detailing specific practices, or were outside the operational scope of CE/decarbonization implementation within I4.0/I5.0-enabled manufacturing systems. Bibliometric papers that documented publication trends without specifying actionable practices were excluded. Subsequent to this screening phase, the dataset was condensed to roughly 553 articles

The third filter phase involved a comprehensive review of the full text, employing more stringent eligibility criteria centered on contextual alignment and applicability. Full texts were evaluated to confirm a focus on enterprise or manufacturing perspectives regarding the implementation of I4.0/I5.0-enabled CE and net-zero practices (Konstantinidis *et al.*, 2023). Only studies that explicitly identified one or more practices that firms or value-chain actors in manufacturing could adopt, implement, or operationalize were retained. Studies were excluded if they referenced relevant themes but focused solely on government policy framing or consumer behavior without explicitly linking to firm-level actions or industrial implementation. After conducting full-text screening, 240 studies met all criteria and were incorporated into the final review corpus. The 240 articles collectively represent the essential literature available as of December 2025 regarding the intersection of I4.0/I5.0, sustainable manufacturing, CE/resource-retention strategies, and net-zero carbon practices. The analytical search was closed on 31 December 2025. Publications carrying a formal 2026 publication date that became available or were consulted during manuscript revision were treated as supplementary post-cutoff references. They were used only to update conceptual interpretation, framework comparison, and discussion of emerging developments and were not incorporated into the PRISMA counts, the 240-study qualitative coding corpus, or the CiteSpace bibliometric networks. Figure 2 synthesizes the complete sequence of identification and screening decisions, adhering to a PRISMA-style flow structure to ensure transparency and replicability.

2.4 Data Extraction Synthetic Analysis:

The systematic synthesis and bibliometric analysis were performed on a final corpus of 240 eligible peer-reviewed articles. Thus, the bibliometric component should be understood as a representation of the implementation-oriented knowledge structure at the convergence of I4.0/5.0, circular manufacturing, CPLM, and net-zero manufacturing, rather than as a comprehensive overview of all publications within these wider research domains. Conceptual articles, purely bibliometric studies, and other records that failed to meet the implementation-oriented eligibility criteria were excluded from the bibliometric networks. This boundary is significant as the resultant thematic clusters, keyword relationships, journal co-citation patterns, and author networks illustrate the intellectual structure of the chosen implementation-oriented corpus.

While collecting the final set of 240 articles, we conducted qualitative content analysis to extract and synthesize appropriate information. Drawing on the findings from the thematic analysis, we undertook an iterative synthesis process to conceptually integrate the identified

themes. A combination of CiteSpace (6.4.R1) software was employed for coding qualitative data and bibliometric analysis, while MATLAB was utilized for aggregating and organizing the coded data, in accordance with the methodology of the foundational study. We employed a mixed inductive-deductive coding approach, ensuring that the coding was rooted in the literature while also being informed by an established conceptual model of circular value chain practices (DeJonckheere *et al.*, 2024).

Two distinct source roles were upheld throughout the review. The analytical review corpus comprised solely peer-reviewed journal articles obtained via a predetermined search strategy in Scopus and Web of Science, meeting the criteria for publication period, language, document type, topical relevance, and implementation orientation. Only studies belonging to this corpus contributed to the PRISMA counts, qualitative practice coding, and CiteSpace bibliometric networks. PRISMA 2020 emphasizes transparent reporting of study identification and selection rather than simply displaying a flowchart (Page *et al.*, 2021). A separate supplementary source set was used for contextual interpretation and conceptual development. This collection comprised policy documents, institutional reports, industry guidance, foundational conceptual publications, and post-cutoff studies referenced during the manuscript revision process. Supplementary sources were utilized to clarify definitions, interpret findings, compare the proposed framework with existing models, and discuss current policy and technological advancements; these sources were excluded from the 240 eligible articles and did not contribute to the bibliometric analysis.

We initially identified the essential elements that follows an explicit mixed inductive–deductive. First, a deductive scaffold was established from prior circular value-chain research, the R0-R9 resource-retention hierarchy, product lifecycle/CPLM literature, established net-zero mechanisms, and the preceding net-zero circular value-chain model. These sources defined the broad analytical dimensions that the framework needed to accommodate without predetermining the final practice categories. Second, the 240 eligible studies were subjected to qualitative coding to identify concrete practices, interventions, capabilities, technologies, and managerial mechanisms relevant to the three research questions. Similar expressions referring to the same underlying activity were consolidated, producing approximately 50 distinct practice codes.

Third, the practice codes were compared iteratively and grouped according to their functional position in the manufacturing lifecycle and their organizational role. Practices directly associated with product and material flows were synthesized into eight primary categories: Circular Eco-Design & Development (CEDD), Sustainable Procurement & Sourcing (SPS), Clean Manufacturing & Process Efficiency (CMPE), Green Distribution & Reverse Logistics (GDRS), Circular Use & Service (CUS), Lifecycle Extension & Refurbishment (LER), Waste-to-Resource Recycling (WRR), and Transparent Sustainability Communication (TSC). Cross-cutting practices that enabled implementation across more than one lifecycle stage were consolidated into five support categories: Strategic & Organizational Net-Zero Vision (SONZ), Green Innovation & Technology Development (GITD), Regenerative Business Models (RBM), Synergistic Collaboration Networks (SCN), and Sustainability Culture & Skills (SCS).

Fourth, each practice was cross-mapped to the R0-R9 strategies, relevant digital capabilities, I5.0 principles, net-zero mechanisms, and selected SDGs. Fifth, category boundaries and duplicated practices were reviewed iteratively to improve internal coherence and align each

practice with the category that most closely reflected its primary organizational function. The bibliometric clusters were used as a complementary triangulation lens to assess whether the conceptual framework covered the principal thematic domains represented in the corpus; the clusters were not used mechanically to generate the eight primary or five support categories. Author discussion was used to resolve ambiguous classifications and refine labels, whereas the substantive categories and practice content remained grounded in the reviewed literature.

The study did not include primary empirical testing. The proposed framework lacked validation through statistical, predictive, or field-based methods. We conducted a SLR to confirm that each linkage or proposition in the model is supported by either conceptual or empirical evidence from prior studies. In this case, our model suggests that I5.0, digital product passports, enabled by digital traceability infrastructures and deployed here in support of I5.0 sustainability, transparency, and lifecycle-governance objectives, enhance recycling (R8) by supplying material information (Jensen *et al.*, 2023). This assertion is supported by sources that examine the relationship between product passports and recycling. The proposed framework was cross-checked to strengthen its alignment with the gaps identified in the introduction, particularly the limited integration between digital and human-centric and circular/net-zero strategies. Each component of the model was examined to determine its contribution to the integration. Insights obtained from an initial iteration of this framework, derived from previous research or peer discussions, were utilized to enhance the clarity of categories and their interrelations. In this regard, the framework was revised to integrate CPLM more distinctly into the value-chain structure and to align I5.0 technologies with specific functional roles rather than treating them as a single, undifferentiated category (Atif, 2023).

Figure 7 presents integrated conceptual framework for a net-zero manufacturing ecosystem, combining I5.0 enablers with circular product lifecycle strategies (R0-R9 loops). Two major inputs I5.0 technologies & human-centric innovation, and CPLM along with R0-R9 strategies across value chain stages converge to form the integrated net-zero manufacturing ecosystem, which delivers sustainable outcomes such as net-zero carbon emissions, resource circularity, and enhanced resilience. This approach was selected to facilitate strategic thinking and research, offering a framework that can be tailored to various manufacturing contexts. The proposed framework was structured to be conceptually thorough while maintaining efficiency: thorough in addressing the entire product lifecycle and key enabling factors, yet simple in consolidating related concepts to minimize complexity for the user.

The methodology encompasses an integrated conceptual development process, grounded in a comprehensive SLR and precise synthesis. The methodology is primarily qualitative and interpretive, characteristics of a theory-building framework. The proposed framework is grounded in existing research and exhibits logical cohesion among its components. The following sections will present the developed theoretical model, accompanied by an analysis of its components and implications. Future empirical studies may apply this model to evaluate its empirical applicability and effectiveness in real manufacturing settings, as discussed later in the research.

SDG alignment was assessed at the level of the intended practice mechanism rather than through broad thematic similarity alone. Each coded practice was compared with the wording and substantive intent of relevant official United Nations SDG targets. An SDG was assigned when

the practice exhibited a direct and defensible mechanism connecting it with at least one relevant target; for instance, material-efficiency and waste-reduction practices were considered in relation to SDG 12, sustainable industrial and technological innovation practices to SDG 9, renewable-energy and efficiency measures to SDG 7, climate-mitigation practices to SDG 13, workforce development and decent-work practices to SDGs 4 and 8, and collaborative ecosystem practices to SDG 17. The two coders independently reviewed the proposed mappings and resolved differences through discussion and consensus. The SDG labels in Table 5 indicate conceptual alignment with official target areas, rather than providing empirical evidence that the implementation of a specific practice has resulted in a measurable SDG outcome.

3. Results

3.1 Bibliometric Analysis Results

A combination of analytical and computational tools was employed to ensure a systematic and rigorous assessment of the dataset. Bibliographic records retrieved from *Scopus and Web of Science* were initially processed in R using the RStudio environment, where data cleaning procedures such as duplicate removal, structuring, and preliminary keyword analysis were performed. CiteSpace (version 6.4.R1) was subsequently used to construct and visualize bibliometric networks, including co-citation, keyword co-occurrence, and collaboration maps, enabling the identification of dominant research themes and intellectual structures.

In parallel, an in-depth qualitative analysis was conducted on the 240 selected articles using CiteSpace for coding and MATLAB for data organization and synthesis. Each study was carefully examined to extract specific practices related to I5.0, sustainable manufacturing, CE, and net-zero strategies. Individual practices such as design for recyclability, renewable energy adoption, and industrial symbiosis were systematically coded and grouped into broader conceptual categories, including collaboration and resource-sharing models. A combined inductive deductive approach, informed by circular value chain theory and grounded in empirical observations, resulted in the identification of approximately 50 distinct practice codes. Each code was consistently linked to relevant articles, ensuring analytical reliability and comprehensive coverage. The integration of these tools enabled a holistic approach, where bibliometric analysis captured the structural evolution of the field, and qualitative coding provided deeper insights into actionable pathways for net-zero manufacturing ecosystems.

The complete clustering solution contained 19 clusters. To maintain interpretive clarity, 15 clusters with sufficient size and thematic coherence were retained for substantive interpretation, whereas four small or weakly differentiated residual clusters were not discussed in detail. Consequently, references to 19 clusters denote the complete algorithmic solution, while references to 15 clusters denote the subset retained for interpretation. This distinction is applied consistently in the methods, results, figure captions, and discussion.

3.1.1 Research Thematic Areas and Clusters

The bibliometric analysis identified several major thematic clusters of research within the I4.0/5.0, sustainable manufacturing, CE and net zero literature. The complete CiteSpace solution contained 19 clusters, of which 15 were retained for substantive interpretation on the basis of their

size and thematic coherence. A science mapping of co-citation and keyword networks revealed 15 distinct thematic areas, of which four were especially prominent are presented in Figure 3.

CE (Cluster #0): This cluster forms the intellectual core of the network, exhibiting the highest density and strongest interconnections with all major themes. It reflects the central role of CE principles in shaping net-zero manufacturing discourse, particularly through resource efficiency, waste valorization, and closed-loop production systems. The prominence of this cluster indicates that circularity is not treated as a standalone concept but as an integrative foundation for redesigning product lifecycles and industrial ecosystems. Its strong links with digitalization and emerging industrial paradigms suggest that CPLM is increasingly positioned as a key pathway toward achieving carbon neutrality.

Sustainable supply chain management (Cluster #1): This cluster highlights the growing importance of integrating sustainability principles across supply chain networks. It focuses on coordination mechanisms, resource optimization, and performance management strategies that enable environmentally and socially responsible operations. The cluster emphasizes resilience, transparency, and efficiency within supply chains, reflecting increasing attention to sustainable sourcing, logistics optimization, and stakeholder collaboration. Its close proximity to the CE cluster suggests that sustainable supply chain management plays a critical role in operationalizing circular practices and facilitating the transition toward net-zero manufacturing ecosystems.

I5.0 (Cluster #2): This cluster represents a rapidly evolving research frontier that emphasizes human-centric, resilient, and sustainability-driven manufacturing systems. Closely connected to the core cluster, it reflects the transition toward integrating advanced technologies with environmental and social objectives. Within the context of net-zero ecosystems, I5.0 is conceptualized as an enabler of intelligent and adaptive production systems that support lifecycle optimization, resource circularity, and low-carbon operations. Its prominence highlights a paradigm shift from purely efficiency-oriented automation to value-driven, sustainable industrial transformation.

I4.0 (Cluster #3): This cluster represents the technological backbone underpinning digital transformation in manufacturing, encompassing smart systems, automation, data analytics, and cyber-physical integration. While relatively mature, it remains highly influential in enabling the transition toward net-zero ecosystems. The cluster's strong interlinkages with CE and I5.0 suggest that I4.0 technologies serve as critical enablers for implementing circular product lifecycle strategies, improving energy efficiency, and supporting real-time decision-making. Its position in the network highlights its foundational role in bridging conventional manufacturing systems with advanced, sustainability-oriented industrial models.

Industrial ecology (Cluster #4): This cluster reflects a systems-oriented perspective on resource flows, environmental impacts, and inter-organizational linkages within industrial systems. It emphasizes the application of ecological principles to industrial activities, including material flow analysis, waste exchange, and industrial symbiosis. The cluster highlights how interconnected production systems can mimic natural ecosystems to enhance resource efficiency and reduce environmental burdens. Its linkage with CE and recycling-related themes indicates that industrial

ecology provides a critical conceptual and analytical foundation for designing integrated, low-carbon, and resource-efficient manufacturing ecosystems.

Beyond these top clusters, the bibliometric mapping indicated up to 19 thematic areas in total, revealing a diverse and mature research landscape. Specialized topics in smaller clusters include, such as, human-centric, recycling, strategic roadmap, food supply chain, carbon trading, electrocatalyst, and sustainable operations. The existence of these niche themes, alongside the major clusters, suggests that CE scholarship has branched out to address circular manufacturing and net zero broad questions and specific technical or contextual issues.

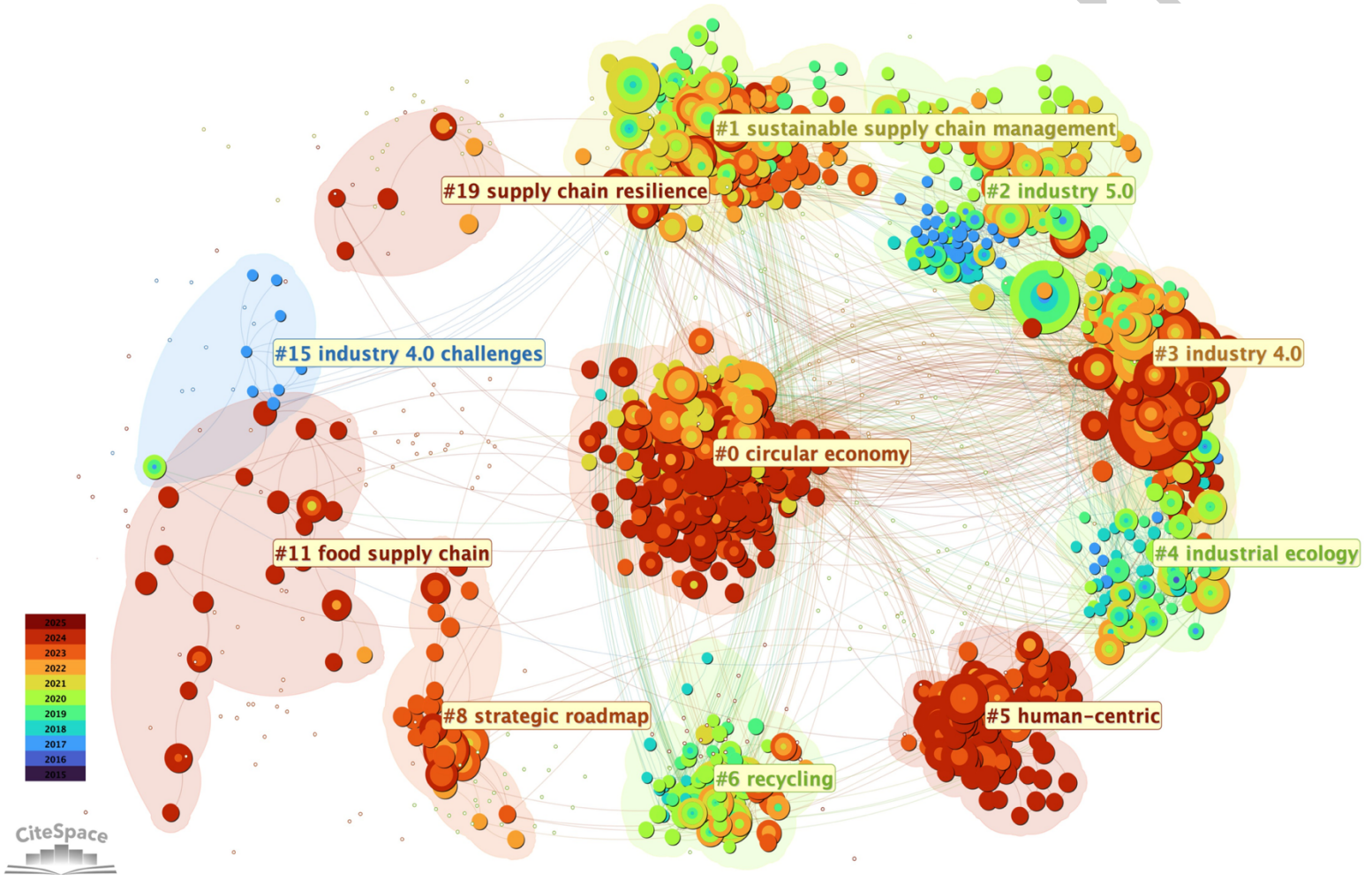


Figure 3: Top research thematic areas in the I4.0/I5.0, circular manufacturing and net-zero literature (2015-2025)

3.1.2 Keyword Co-Occurrence Network

The keyword co-occurrence network illustrated in Figure 4 offers a comprehensive view of the knowledge structure underpinning research on net-zero manufacturing ecosystems. By mapping the frequency and interconnections of key terms, the network reveals how the field has

evolved from technology-driven discussions toward integrated sustainability and decarbonization frameworks. The visualization highlights both the conceptual core and the diversification of themes that collectively shape this research domain. At the center of the network lies a dense concentration of foundational keywords, including “*decarbonization*,” “*carbon emissions*,” “*carbon neutrality*,” “*climate change*,” and “*sustainable development*.” These core terms indicate that the primary focus of the literature is on reducing environmental impact and transitioning toward low-carbon industrial systems. Their strong interlinkages suggest that decarbonization is not treated in isolation but is deeply embedded within broader sustainability and policy-oriented discussions. The presence of geographically tagged nodes such as “China” further reflects the empirical emphasis of the literature, particularly in regions actively pursuing carbon reduction strategies.

Radiating outward from this core, several interconnected thematic domains emerge. A prominent cluster is associated with digital and industrial transformation, where keywords such as “*I4.0*,” “*I5.0*,” “*digital twin*,” “*internet of things*,” “*machine learning*,” and “*artificial intelligence*” are highly interconnected. This indicates that advanced digital technologies are increasingly recognized as critical enablers of net-zero manufacturing. The close proximity of these terms to sustainability-related keywords suggests a convergence between digitalization and environmental performance, reinforcing the role of smart systems in optimizing resource use and reducing emissions across product lifecycles.

Another significant cluster relates to energy systems and optimization, characterized by terms such as “*renewable energy*,” “*energy consumption*,” “*energy efficiency*,” “*hydrogen storage*,” “*solar power generation*,” and “*optimization*.” This grouping highlights the operational dimension of net-zero transitions, where energy management, system efficiency, and cost-effectiveness are key research priorities. The frequent co-occurrence of “*optimization*,” “*cost reduction*,” and “*performance assessment*” indicates a strong emphasis on balancing environmental objectives with economic feasibility, particularly in the context of sustainable industrial operations. In addition, the network reveals specialized clusters focused on material processing, circular resource use, and industrial applications. Keywords such as “*recycling*,” “*battery recycling*,” “*metal recovery*,” “*construction materials*,” and “*additive manufacturing*” reflect the growing importance of CPLM. These terms illustrate how resource recovery, material efficiency, and lifecycle extension strategies are integrated into the broader net-zero agenda. Similarly, clusters related to “*wastewater treatment*,” “*bioremediation*,” and “*waste management*” indicate the expansion of sustainability considerations into environmental remediation and industrial ecology.

The keyword co-occurrence structure demonstrates that research in this field has progressed toward a highly interconnected and interdisciplinary landscape. Rather than focusing solely on isolated technological or environmental aspects, recent studies integrate digital innovation, energy systems, circularity, and policy dimensions into a unified framework for net-zero manufacturing. The network thus reflects a shift toward ecosystem-level thinking, where achieving carbon neutrality depends on the coordinated interaction of technologies, processes, and sustainability strategies across the entire product lifecycle.

A second layer of influential outlets includes journals focused on energy systems and applied sustainability, such as *Applied Energy*, *Energy*, and *Energy Economics*. These journals contribute significantly to discussions on energy optimization, decarbonization pathways, and techno-economic analysis, highlighting the operational dimension of achieving net-zero manufacturing. Their strong co-citation ties with sustainability-oriented journals suggest a convergence between environmental objectives and energy system innovation.

In addition, the network reveals the presence of interdisciplinary journals such as *Science of the Total Environment*, *PLOS ONE*, and *Nature*, which extend the research scope to environmental impacts, ecological systems, and broader scientific inquiries. This indicates that net-zero manufacturing is not confined to industrial or managerial perspectives but is increasingly examined within global environmental and scientific contexts. Similarly, journals like *Building and Environment* and *Journal of Industrial Ecology* reflect the integration of lifecycle thinking, urban sustainability, and resource flow analysis into manufacturing research.

The distribution of leading journals demonstrates that scholarship in this domain is concentrated within a set of high-impact, sustainability-focused outlets while simultaneously drawing from energy, engineering, and environmental science disciplines. The co-citation structure underscores the evolution of the field toward a holistic and systems-oriented perspective, where achieving net-zero manufacturing ecosystems requires the integration of technological innovation, energy transition, and CPLM across multiple research domains.

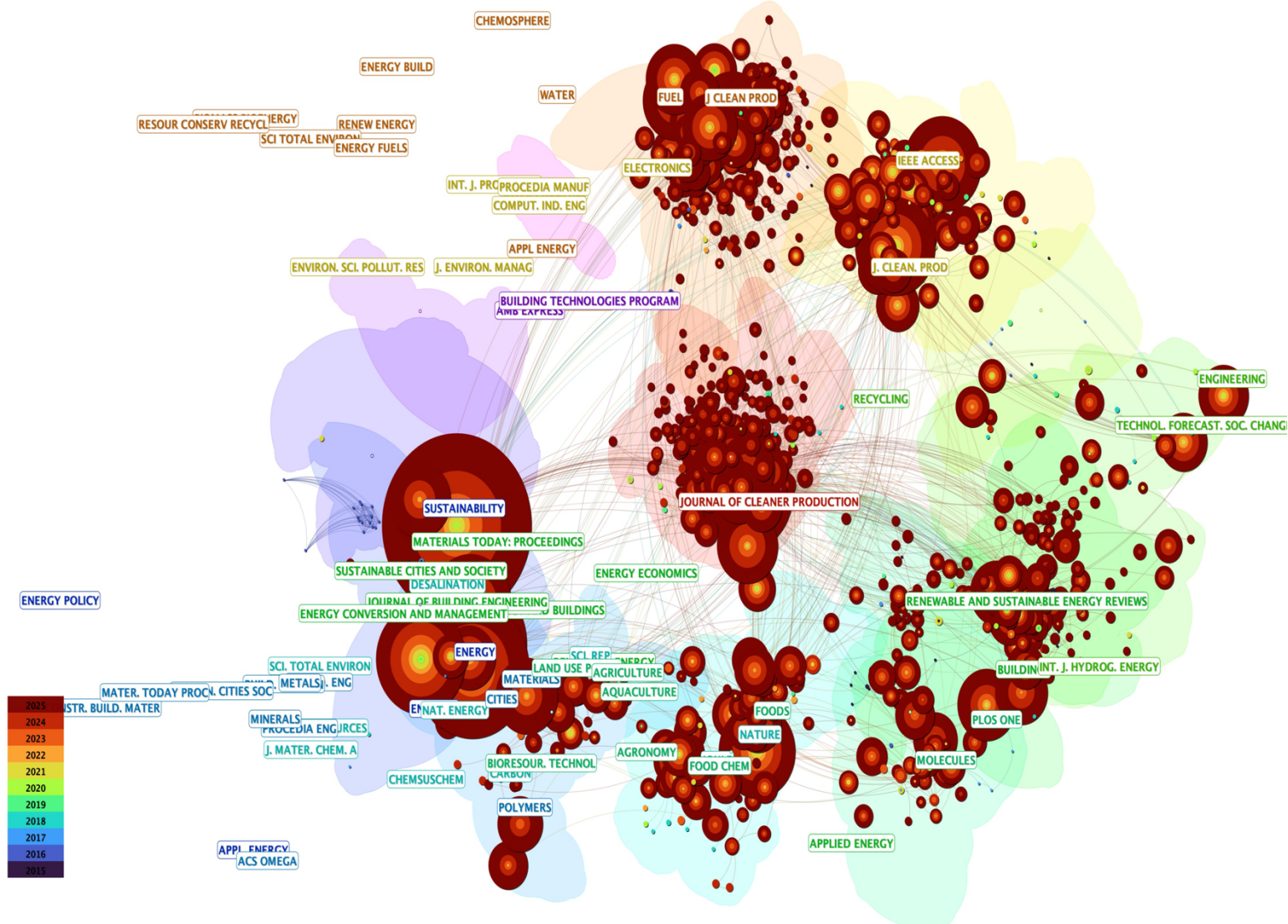


Figure 5: Journal co-citation network illustrating leading publication outlets and their intellectual linkages in I4.0/5I.0, circular manufacturing and net-zero research

3.1.4 Research Author Contribution and Collaboration Network

Figure 6 presents the author co-citation and collaboration network, offering insights into the intellectual contributors shaping research on net-zero manufacturing, I5.0, and CPLM. The network reveals a multi-clustered structure, where groups of authors form distinct yet interconnected research communities, reflecting both thematic specialization and collaborative knowledge development.

At the center of the network, a dense cluster of frequently cited authors indicates a strong concentration of influential contributions. Scholars such as *Chobakhlo*, *Ribeiro da Silva*, *Dwivedi*

YK, Perona M, and Kumar V appear prominently, suggesting their significant role in advancing discussions related to digital transformation, sustainability integration, and industrial system optimization. The tight interconnections within this cluster highlight a shared intellectual foundation, where research on smart manufacturing, decision-making frameworks, and sustainability performance converges.

One prominent group includes researchers such as *Gawankar SA, Kivikyto-Reponen P, and Garza-Reyes C*, whose work is closely associated with sustainable operations, supply chain resilience, and CE practices. Their positioning indicates a strong contribution to operationalizing sustainability within manufacturing systems. Another cluster, featuring authors such as *Wang L, Petridis A, and Gajdzik B*, reflects growing interest in advanced technologies and their application to industrial sustainability, particularly in the context of I4.0 and I5.0 integration.

In addition, the network highlights contributions from authors working on environmental systems and resource management, including *Seliger G, Ulgiati S, and Seppälä J*. This group emphasizes lifecycle assessment, resource efficiency, and ecological impacts, reinforcing the importance of environmental metrics in net-zero transitions. A separate cluster involving *Ayala NF, Sarkis J, and Kumar S* illustrates the intersection of supply chain management, circularity, and sustainability strategy, indicating the role of logistics and value chain redesign in achieving low-carbon ecosystems.

The author collaboration structure demonstrates that the field is shaped by a diverse and interconnected group of scholars spanning engineering, management, environmental science, and operations research. The co-citation patterns suggest a gradual convergence of these disciplines toward a unified research agenda focused on designing net-zero manufacturing ecosystems. This collaborative and interdisciplinary knowledge base is critical for advancing integrated solutions that combine technological innovation, CPLM, and sustainability-driven industrial transformation.

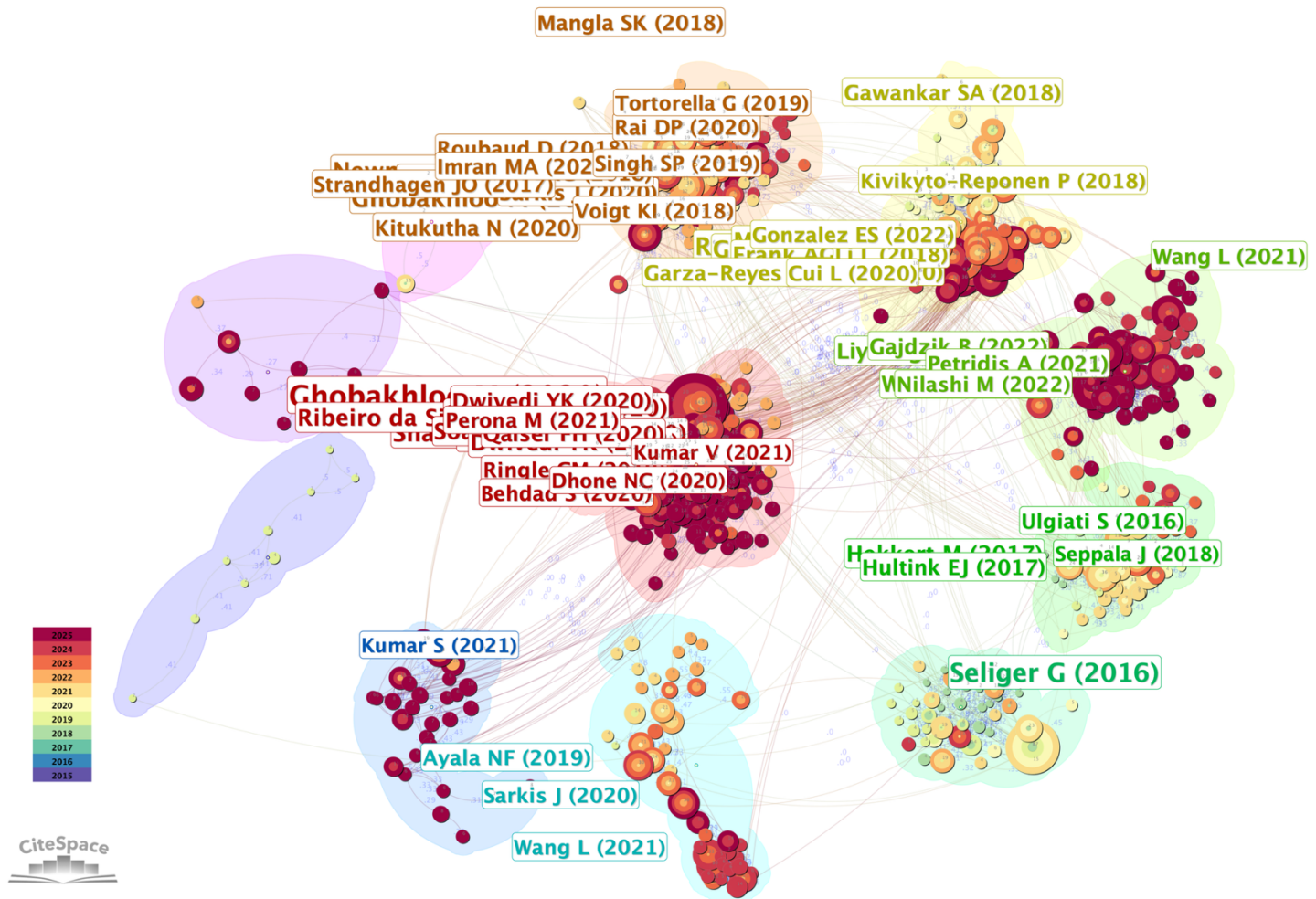


Figure 6: Author co-citation and collaboration network showing key contributors and research communities in net-zero manufacturing and circular systems.

Next, we organized these coded practices into higher-level categories using both established frameworks and emerging themes. Deductively, we drew on Porter's value-chain segmentation and recent I4.0/I5.0 literature to inform our structure. Inductively, we looked at how practices naturally clustered by theme such as design/innovation actions became a "Circular Eco-Design & Development" group, while energy-related practices formed "Clean Manufacturing & Process Efficiency". We then refined the categories by aligning them with both the traditional (linear) and circular value-chain views, explicitly adding a net-zero dimension. The final framework illustrated in Figure 7 includes primary value-chain categories Circular Eco-Design & Development (CEDD), Sustainable Procurement & Sourcing (SPS), Clean Manufacturing & Process Efficiency (CMPE), Green Distribution & Reverse Logistics (GDRS), Circular Use & Service (CUS), Life-cycle Extension & Refurbishment (LER), Waste-to-Resource Recycling (WRR), and Transparent Sustainability Communication (TSC) as well as supporting enablers Strategy & Org Net-Zero Vision (SONZ), Regenerative Business Models (RBM), Green

Innovation & Tech Dev (GITD), Synergistic Collaboration Networks (SCN), and Sustainability Culture & Skills (SCS).

Finally, Table 5 presents the consolidated list of 50 Integrated I5.0, circular PLM and net-zero practices. Each row shows the practice code, a description, the associated R-loop(s), and applicable UN SDGs 3,4,7,8,9,10,11,12,13, and17. This structured integrates the I5.0, CPLM, and decarbonization perspectives into a coherent framework. By explicitly aligning each practice with an SDG, our model highlights how circular-net-zero strategies contribute to broader sustainability targets such as SDG 12 on responsible production and SDG 13 on climate action. This approach extends prior CE frameworks by embedding net-zero objectives and SDG alignment into every dimension of the value chain.

This study presents a multilayer net-zero manufacturing ecosystem that integrates I5.0 with a circular product lifecycle model, as illustrated in Figure 7. It demonstrates a cohesive framework that combines CE loops, enabling elements of I5.0, and net-zero principles within a manufacturing context. The model prioritizes human centricity within the decarbonization agenda, thereby redefining the concept of manufacturing excellence. The model posits that labor, safety, inclusion, and wellbeing should not be regarded as secondary concerns, but rather as essential conditions for sustainable industrial performance. This perspective is consistent with the principles of II5.0, which advocate for future production systems to transcend solely automation-driven or profit-centric approaches. Instead, it emphasizes the integration of technological progress with resilience, stakeholder value, and adherence to planetary limits. The model indicates that net-zero manufacturing encompasses not only a technical transition in machinery and energy systems but also an organizational transformation regarding value creation, worker interaction with technology, and the role of industrial systems in serving society.

The model reveals that net-zero performance is allocated throughout the entire product lifecycle, encompassing early co-creation and design, production, delivery, usage, and end-of-life recovery. The lifecycle logic of the figure indicates that emissions and waste are influenced by factors beyond the confines of the factory gate. Upstream factors include design choices, material selection, and forecasting quality, while downstream factors pertain to the potential for products to be repaired, reused, remanufactured, or recycled. This interpretation is substantiated by research on climate and circularity, indicating that industrial decarbonization necessitates comprehensive value-chain engagement and that circular strategies can markedly decrease emissions from essential material systems by prolonging the use of products, components, and materials. The reverse-flow pathway in the model functions as a fundamental net-zero mechanism, as it diminishes the demand for virgin materials, reduces embodied carbon, and transforms waste management into a productive cycle of resource recovery.

The digital layer of the model illustrates the practical functioning of such an ecosystem. Technologies including IoT, AI, machine learning, edge connectivity, digital twins, cloud platforms, and blockchain-based traceability function collectively as an integrated intelligence infrastructure throughout the lifecycle. Recent research emphasizes the significance of digital twins due to their role in real-time synchronization, feedback mechanisms, and lifecycle integration. Additionally,

digital product passport frameworks enhance transparency by documenting information related to environmental performance, resource utilization, repairability, recyclability, and end-of-life management. From a net-zero perspective, digitalization is crucial when it enhances the visibility and actionability of environmental performance: energy optimization can occur sooner, defects can be averted prior to material loss, circular routes can be implemented more swiftly, and sustainability claims can be linked to verifiable lifecycle data instead of vague corporate commitments.

The governance and outcome layers indicate that the model encompasses not only intelligent production but also coordinated industrial citizenship. The model posits that net-zero manufacturing relies on the integration of policymakers, regulations, enterprises, communities, consumers, technology partners, and the environment within a unified framework. It emphasizes the necessity of aligned institutions, interoperable standards, transparent reporting, and socially inclusive capability building. This aligns with contemporary net-zero guidance that highlights value-chain accounting and effective communication, as well as climate assessment literature indicating that industrial transition necessitates coordinated policy, emerging market demand for low-emission products, infrastructure investment, and workforce learning and retraining. The primary contribution of the model lies in its integrated logic, wherein workforce excellence, organizational resilience, societal wellbeing, circular resource efficiency, and emissions reduction are framed not as distinct agendas, but as interdependent outcomes within a singular multilayer manufacturing ecosystem.

Net-Zero Manufacturing Ecosystems, Industry 5.0 and Circular Product Lifecycle Management (Co-Creating Sustainable Value through Intelligence, Resilient & Circular System)

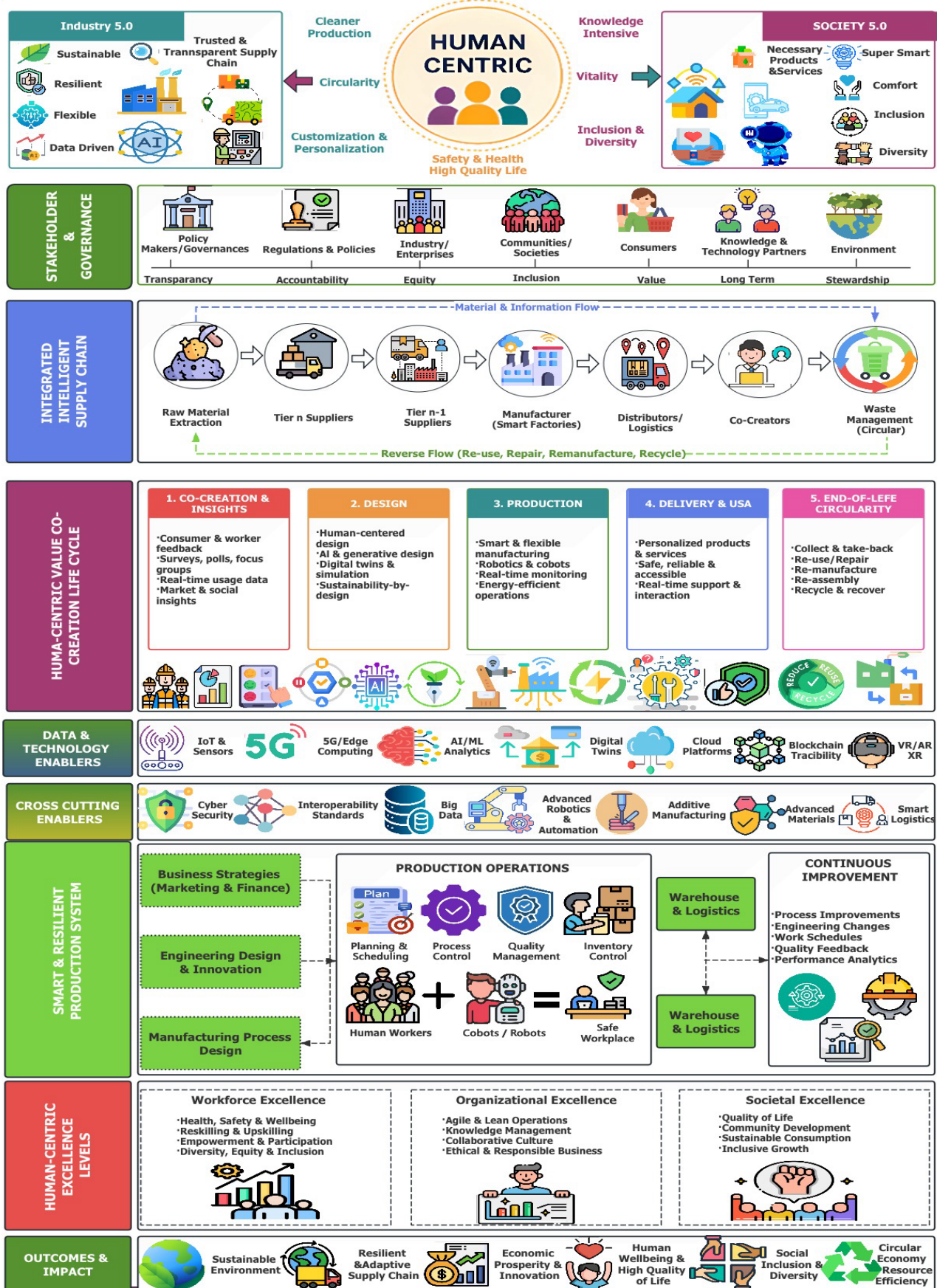


Figure 7: Multilayer net-zero manufacturing ecosystems integrating I5.0 and Circular Product Lifecycle (authors own drawing)

Figure 7 illustrates conceptual relationships supported by literature and anticipated implementation pathways, rather than empirically derived causal effects. The arrows and outcome domains illustrate how the reviewed literature indicates that the identified practices may enhance circularity, decarbonization, resilience, and human-centric performance when executed under suitable organizational and technological conditions. These should not be construed as quantified effect sizes, causal coefficients, or as evidence that each practice yields the specified outcome in all manufacturing contexts.

Table 3 outlines the key value chain categories within the ecosystem, highlighting their correspondence with circular R-strategies and providing examples of applied I5.0 technologies. Subsequently, Table 4 provides a summary of the support categories (enablers), their functions within the framework, and their relationship to the principles of I5.0.

Table 3: Circular value chain categories in net-zero manufacturing ecosystems with R-strategies and I5.0 enablers

Value Chain Category	Aligned Circular Strategies (R#)	I5.0 Digital Enablers and I5.0-Oriented Application	Applications Description
Circular Design & Development (CEDD)	R0 Refuse (eliminate unnecessary parts), R1 Rethink (alternative functions/business models), R2 Reduce (material and energy efficiency in design).	AI-driven eco-design tools facilitate lightweighting and material optimization. Generative design software enhances creative solutions. Material databases provide sustainability data. Collaboration platforms enable co-design with customers, emphasizing a human-centric approach.	The product is designed with a modular architecture that facilitates component replacement and future upgrades, thereby supporting R5 refurbishment and, where the product is restored to as-new functional specifications through a controlled industrial process, R6 remanufacturing
Sustainable Procurement & Sourcing (SPS)	R0 Refuse (avoid toxic/unsustainable materials), R2 Reduce (source recycled content to reduce virgin use), R1 Rethink (supplier partnerships for servicing, such as	Utilization of blockchain technology to enhance supply chain transparency regarding the provenance of recycled or certified materials, implementation of	Emphasizing the use of recycled or bio-based materials in the legislation of Materials; employing blockchain technology to authenticate the content of recycled materials, thereby indirectly

	supplier takes back packaging).	supplier data exchange platforms, and application of AI for the analysis of sustainability risks associated with suppliers.	supporting R8 at the source; and organizing supplier-managed systems for packaging reuse, in accordance with R3.
Clean Manufacturing & Process Efficiency (CMPE)	R2 Reduce (minimize scrap, optimize energy use), R3 Reuse (reusing process inputs, like water, heat), R4 Repair (maintenance of equipment to avoid breakdown waste).	IoT sensors facilitate real-time monitoring of equipment and resource consumption. A digital twin of the production line enables the simulation of efficiency improvements. Predictive maintenance systems, utilizing AI, aim to minimize downtime and defects. Autonomous robots contribute to waste reduction in assembly processes.	Implement closed-loop cooling water systems that reuse water in processes, align with R3, utilize real-time energy management to reduce energy waste, and employ predictive maintenance to avert quality defects that may lead to scrap, in accordance with R2.
Green Distribution & Reverse Logistics (GDRS)	R2 Reduce (efficient logistics to cut fuel/emissions), R3 Reuse and R8 Recycle (establishing reverse logistics for returns, packaging reuse, and recycling flows).	IoT-enabled fleet management optimizes routes and load factors. Electric or alternative-fuel delivery vehicles represent sustainability technology in I5.0. Warehouse automation facilitates the handling of returned products through robotic sorting. Tracking systems, potentially utilizing blockchain, are employed for product returns and core identification in remanufacturing.	Employing AI to enhance delivery routes for fuel minimization (R2 carbon reduction), implementing a take-back program wherein IoT-connected products indicate end-of-life and initiate pickup (facilitating R5-R8), and reusing delivery packaging on multiple occasions (R3).

Circular Use and Service (CUS)	R3 Reuse (facilitating secondary use), R4 Repair (maintenance services), R1 Rethink (product-as-a-service or sharing models to increase utilization).	Remote sensing of products through IoT facilitates predictive maintenance. Augmented reality guides assist repair technicians and customers in performing repairs efficiently. Platforms enable sharing and management of products as a service, allowing for the scheduling of product use among multiple users. Digital contracts, or smart contracts, support leasing models.	Providing products through leasing or pay-per-use models, facilitated by a digital platform, aligns with R1. IoT sensors notify users of maintenance needs prior to machine failure, as indicated in R4. Additionally, the use of AR headsets by field technicians enables efficient on-site repairs of complex machinery, contributing to prolonged equipment lifespan (R4).
Lifecycle Extension & Refurbishment (LER)	R5 Refurbish, R6 Remanufacture, R7 Repurpose (core activities in this stage). Also supports R3 (because refurbished products are reused) and R2 (reduces need for new production).	Reverse supply chain management systems involve inspection and grading through AI and computer vision to categorize returned products based on their condition. Robotics are utilized for the disassembly of products, ensuring safe and efficient component harvesting. Knowledge databases, potentially enhanced by AI, are employed to optimize the remanufacturing process. Additionally, 3D printing is applied to create spare parts for obsolete components, facilitating refurbishment efforts.	A factory refurbishment process for used smartphones: devices are collected, automatically assessed by AI for damage, then repaired or have modules replaced, and sold as certified refurbished units (R5). A remanufacturing line for industrial equipment, where returned cores are disassembled by robots, critical parts are re-machined or replaced, and the product is reassembled with like-new warranty (R6).
Waste-to-Resource	R8 Recycle (materials recycling processes),	Advanced sorting technologies,	A recycling facility that utilizes AI-based

Recycling (WRR)	R9 Recover (energy recovery for non-recyclables). Also, R7 Repurpose in cases where parts/materials are directly repurposed into new products without full reprocessing.	including AI-driven optical sorters for materials, IoT applications in recycling facilities for tracking material flows, and chemical recycling technologies for plastics represent significant technological innovations. Additionally, industrial symbiosis platforms that digitally match one firm's waste to another's input facilitate repurposing. Emissions control and energy recovery systems with real-time monitoring enhance efficiency when R9 is implemented.	sensors to sort manufacturing scrap by alloy, subsequently recycling it in-house to produce new raw materials, thereby achieving internal loop closure (R8). Employing waste heat from one process to energize another demonstrates the repurposing of energy flows. An AI marketplace facilitates the identification of buyers for waste by-products, such as fabric scraps, which are sold to insulation manufacturers, exemplifying the repurposing of materials.
Transparent Sustainability Communication (TSC)	Supports R1-R9 by providing necessary info (R3 reuse needs info on product specs for second-hand users, R8 recycle needs material info). Not an R in itself but crucial for all loops.	Digital product passports, potentially utilizing blockchain technology for enhanced trust, will contain lifecycle data. Customer applications or interfaces will display product footprints along with reuse and recycling options. Cloud-based data sharing systems will facilitate connections among manufacturers, recyclers, and consumers to promote transparency. Additionally, QR codes or IoT tags will be affixed to products	Implementing QR codes on products that, upon scanning, provide users with information on return or recycling procedures can enhance recycling rates. Disseminating carbon footprint and circularity metrics of products to business customers through an online dashboard can enhance trust and potentially encourage customers to return products at the end of their lifecycle.

		to enable scanning and access to recycling instructions.	
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Table 4: Integrated framework support and enabling categories link organizational capabilities, I5.0 concepts, and circular net-zero ambitions.

Support Category	Role in Framework	I5.0 / Organizational Enablers	Illustrative Actions
Strategic & Organizational Net-Zero Vision (SONZ)	Offers leadership commitment and incorporates objectives into fundamental business strategy; establishes targets (zero waste, net-zero by year X) and distributes resources accordingly. Facilitates interdepartmental alignment to achieve CE and emissions objectives.	Establishment of a robust leadership mandate for sustainability, emphasizing I5.0's human-centric values at the leadership level. Implementation of sustainability Key Performance Indicators (KPIs) within performance management, supported by digital tracking mechanisms. Formation of cross-functional governance teams, such as a CE steering committee.	The company establishes a formal Net-Zero 2030 roadmap and a CE strategy, which has received endorsement from the CEO. Executives' bonuses are linked to the attainment of material recycling rates or emission targets, thereby ensuring accountability. Additionally, there is a public commitment to science-based targets and SDGs that inform innovation efforts.
Green Innovation & Technology Development (GITD)	This activity supports continuous improvement and innovative solutions for circular net-zero issues. Develops new materials that are biodegradable and recyclable, implements new process technologies focused on low-carbon methods, and utilizes I5.0 technologies, including AI, for the	Research and development initiatives centered on sustainable technology, collaborative innovation with startups or research institutions, and digital innovation laboratories employing emerging technologies (such as AI, IoTs, and 3D printing) to prototype CE solutions. Facilitates an I5.0	Investing in developing a recyclable version of a currently hard-to-recycle product component; piloting AI for life-cycle assessment during design to quantify footprint quickly for each design iteration; collaborating with tech providers to develop robotics for automated disassembly in the factory.

	discovery of innovative materials.	framework by ensuring that technology is aligned with sustainability objectives.	
Regenerative Business Models (RBM)	Transforming the method of value delivery: shifting from product sales to service provision, creating frameworks in which the company maintains ownership (and consequently responsibility) of products, thereby promoting reuse and recycling. Generates revenue through circular services such as maintenance, take-back, and refurbishment sales, thereby supporting net-zero initiatives.	servitization platforms facilitate the management of subscriptions and leasing arrangements. IoT-enabled usage tracking allows for charging based on actual usage outcomes. Smart contracts are utilized for transactions involving product sharing. Aligns with I5.0 personalization by providing customers with precisely what they require, no excess.	The introduction of a Product-as-a-Service model, specifically offering lighting as a service instead of selling lightbulbs, enables the company to optimize bulb lifespan and facilitate their return for recycling, thereby aligning profitability with CE principles. Establishing a deposit system for products to promote their return. Utilizing usage data from the IoT to charge industrial clients based on output, such as implementing a "pay per printed page" model rather than selling printers outright, incentivizes providers to enhance the longevity and energy efficiency of printers, as maintenance and energy expenses directly impact their profit margins.
Synergistic Collaboration Networks (SCN)	Recognizes that a single company cannot independently close all loops; promotes the flow of materials and information across networks, including	Digital collaboration platforms, such as shared databases and blockchain consortia for material tracing, facilitate the exchange of information. The I5.0	Participating in an industrial symbiosis network in which the waste of one firm is utilized as feedstock by another, exemplified by a chemical plant

	suppliers, distributors, recyclers, governments, and even competitors for shared recycling infrastructure. Improves the resilience and scalability of the circular ecosystem.	Community of Practice focuses on sharing best practices related to human-centric sustainability. Public-private partnerships are established to advance recycling technology. Additionally, standardization efforts aim to align data standards for product passports. Highlights the significance of collective well-being in I5.0, moving beyond mere competition.	employing CO₂ emissions from a cement factory. Collaborating with governmental entities and other companies to invest in a local facility dedicated to the recycling of challenging materials. Distributing an accessible database of component designs to support third-party repairs and remanufacturing efforts. Engaging in policy discussions to influence extended producer responsibility regulations that facilitate circular objectives.
Sustainability Culture & Skills (SCS)	Promotes a culture of circularity and ongoing enhancement within the organization's workforce. Provides training to employees in technical skills, such as data analytics for sustainability and eco-design techniques, as well as in change management to facilitate innovation. Involves employees across all levels in the generation of ideas aimed at waste reduction and enhancing efficiency. Increased	Workforce training programs that integrate digital upskilling with continuous education concepts, the inclusion of frontline workers' input through digital suggestion systems or workshops, change management initiatives employing gamification via applications to promote sustainable behavior on the shop floor, and a recognition and reward system for team achievements in sustainability.	Regular training on I5.0 and CE is essential for teaching factory operators to interpret IoT data dashboards related to energy consumption, thereby enabling them to propose optimizations. Establishing a "Green Champion" team at each facility to oversee material waste and spearhead initiatives, such as a competition aimed at attaining the highest recycling rate within a quarter. It is essential to train design engineers in circular

	engagement correlates with greater success in the implementation of new practices.		design principles and the application of life-cycle assessment (LCA) software.
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This study offers a systematic framework for understanding how circular product lifecycle stages, influenced by R-strategies, effectively supported by I5.0 technologies and human-centric practices, governed by net-zero sustainability objectives. This highlights specific leverage points (categories) and enabling factors (technologies, skills, collaborations) that can support managers in implementation or assist researchers in identifying areas for further exploration. The following section analyzes the particulars of the category and practice of design, understanding the operationalization of each category through specific practices and their interconnections, thus offering a practice-focused perspective that extends the conceptual model.

3.2 Circular Value Chain Categories and Practices

This section translates the theoretical model into actionable guidance by presenting a detailed category and practice outcomes. It aligns each segment of the manufacturing value chain and support system with the corresponding circular strategies (R0-R9) and I5.0 technologies. The objective is to connect the overarching model with a detailed view of implementation: identifying specific practices or initiatives within each category and assessing their contributions to circularity and net-zero outcomes. This exposition offers a practical framework for practitioners and a systematic agenda for researchers to examine the integrated approach. Building upon Table 3, each value chain category is here described with representative practices and how they enact the R-strategies, enhanced by I5.0 capabilities:

3.2.1 Circular Eco-Design & Development (CEDD)

This category involves decisions that significantly influence future sustainability outcomes. Practices involve the implementation of Design for X principles, which include considerations for the environment, reuse, and disassembly. Design for disassembly facilitates the straightforward disassembly of products at the end of their life cycle, thereby supporting R4-R7, as components can be replaced, upgraded, or repurposed. Companies such as Philips has effectively implemented modular design in their lighting, facilitating the reclamation and refurbishment of components (Brock *et al.*, 2019). Another practice involves conducting Life Cycle Assessment (LCA) during the design phase to quantify potential environmental impacts and identify opportunities for design improvements, a process expedited by AI-driven LCA. I5.0 technology, specifically generative design algorithms powered by AI, can generate designs that are lighter or simpler, thereby utilizing less material while still fulfilling specified requirements (Sai *et al.*, 2024). Digital twin prototyping facilitates virtual assessments of durability and reparability, enabling designers to simulate potential product failures and improve the accessibility of essential components for repair (R4). The human-centric approach at this stage may include co-innovation with customers through

crowdsourcing platforms or virtual reality co-design sessions to reassess product requirements, such as investigating service models in lieu of traditional product ownership.

3.2.2 Sustainable Procurement & Sourcing (SPS)

This category encompasses essential practices, including the development of a sustainable procurement policy that mandates specific percentages of recycled or certified sustainable content in supplies. This approach aligns with R8 by fostering demand for recycled materials and R2 by reducing the reliance on virgin inputs. Firms can establish a supplier code of conduct that focuses on carbon footprint reduction and CE principles, while collaborating with suppliers on resource efficiency initiatives (Kazancoglu *et al.*, 2021). I5.0 technology utilizes supply chain mapping tools that employ blockchain to authenticate the origin and recycled content of materials. AI can assist in identifying alternative materials that exhibit reduced environmental footprints. A human-centric approach encompasses collaborative relationships with suppliers, such as the joint development of take-back schemes in which suppliers reclaim packaging or products at the end of their life cycle (Greif *et al.*, 2025). This practice is observed in the electronics and automotive industries as part of extended producer responsibility, specifically in the realms of packaging reuse and material recycling. Vendor-managed packaging reuse involves the utilization of IoT-tracked reusable containers for parts deliveries, thereby decreasing the reliance on single-use packaging. R0 refuse disposable packaging (Li and Wang, 2023).

3.2.3 Clean Manufacturing & Process Efficiency (CMPE)

This category involves lean and sustainable practices within this context. This entails the utilization of lean manufacturing methodologies, including just-in-time, 5S, and Kaizen, which fundamentally reduce waste, in conjunction with the adoption of Six Sigma and quality management strategies to lower defect rates and minimize the quantity of scrapped components. Many factories are adopting closed-loop process systems, including the capture and recycling of water and the reuse of process heat, thus promoting resource reuse in their operations (Hagelüken and Goldmann, 2022). I5.0 technology employs IoT sensors on machinery to collect data for the purpose of optimizing energy consumption and minimizing idle operation. A prevalent approach involves the utilization of a Manufacturing Execution System (MES) equipped with real-time dashboards for monitoring resource consumption. This facilitates the identification of anomalies, such as air compressor leaks that result in energy wastage, allowing for timely intervention by managers and workers (R2) (Jaskó *et al.*, 2020). AI-driven analytics can recommend modifications to process parameters to achieve optimal efficiency, such as altering oven temperature profiles to decrease energy consumption while preserving quality. The human-centric component enables operators to propose suggestions, underpinned by data transparency. Moreover, a digital suggestion system can gather employees' proposals for the reduction of particular waste streams, enabling management to execute these modifications swiftly and promoting a culture of ongoing enhancement. Toyota, noted for its lean methodologies, has instituted energy monitoring for each production cell, enabling teams to engage in competition to reduce energy consumption per unit (Carignano and Costa-Castelló, 2023). These workforce engagement methods align with the collaborative principles of I5.0 and facilitate outcomes that minimize resource consumption.

3.2.4 Green Distribution & Reverse Logistics (GDRS)

Forward logistics practices include route optimization to reduce fuel consumption and emissions, load consolidation, and the implementation of low-carbon transportation methods, such as electric trucks and the preference for rail or sea transport over air freight. Reduction (R2) is achieved by eliminating unnecessary transportation and adopting fuel-efficient methods. In reverse logistics, organizations establish mechanisms that facilitate the return of used products by customers, such as offering return shipping labels or designated drop-off sites, including e-waste bins or trade-in programs for mobile devices (Wu *et al.*, 2023). I5.0 technology employs advanced logistics management systems that utilize real-time traffic and demand data, driven by AI, to optimize delivery planning and minimize travel distances. Geospatial analytics can optimize distribution networks to reduce distances between return points and refurbishment centers, thereby enhancing the efficiency of R5/R6 by effectively supporting these loops (Jeong *et al.*, 2021). Blockchain technology can maintain the integrity of the chain of custody for returned items, particularly when products involve sensitive data or hazardous materials that require certified handling. A good illustration is H&M's garment collection program, which includes in-store drop-off points for convenience in returns and utilizes logistics to transport textiles to sorting and recycling facilities (Brun, 2020). The I5.0 framework integrates sorting robots and near-infrared sorting technology for textiles, thereby improving recycling processes via digital innovation. Distribution is strongly connected to service models; in the context of leased products, it involves the exchange of units for maintenance purposes. The system must efficiently manage both forward (deployment) and reverse (take-back) flows.

3.2.5 Circular Use and Service (CUS)

This value chain category emphasizes the importance of extending product lifespan and maximizing utilization. Preventive and predictive maintenance services, often incorporated with the product or included in service agreements, improve product efficiency and foster energy efficiency during operation, consequently lowering emissions. Numerous manufacturing companies are transitioning from product-based models to service-oriented approaches, exemplified by Rolls-Royce's power by the hour initiative for aircraft engines (Burgio, 2024). This model allows the company to retain ownership and maintenance responsibilities for the engines, thereby incentivizing the production of durable and easily repairable engines, resulting in a circular benefit. This model involves IoT sensors in engines transmitting data to Rolls-Royce, which then performs timely maintenance, illustrating a fundamental application of I4.0/I5.0 with circular outcomes. Moreover, another practice entails the facilitation of secondary markets for products. A firm establishes a certified pre-owned program identical to those employed by numerous electronics and equipment manufacturers, allowing one user to return a product for servicing and subsequent redistribution to another user (R3 direct reuse). An online platform can enhance the alignment of supply and demand for second-hand goods. Augmented reality (AR) remote assistance within the framework of I5.0 technology is increasingly being employed. This enables an expert to support a local technician or user in product repairs via augmented reality overlays, thereby improving repair success rates and reducing the need for in-person visits, which also aids in decreasing carbon emissions (Marino *et al.*, 2024). The human-centric approach entails the design of user interfaces and support systems that facilitate sustainable product usage (Grandi *et al.*, 2025). Likewise, it encompasses offering eco-driving feedback in vehicles and employing printers that display the environmental impact of printing, thus promoting double-sided printing and motivating users to embrace sustainable practices. User engagement is essential for attaining

a net-zero circular ecosystem, requiring participation in activities like product returns and the acceptance of remanufactured components.

3.2.6 Lifecycle Extension & Refurbishment (LER)

This category is defined subsequent to the application of a product or component. Practices involve the creation of refurbishment centers or remanufacturing lines. The implementation of a core management system entails the systematic tracking of returned used products, known as cores in remanufacturing sectors, through a database that serves as an extension of PLM. This approach can help route these products more efficiently to appropriate processing locations. I5.0 technology emphasizes the importance of automation, exemplified by the implementation of robotic disassembly cells in the electronics sector (Shukla *et al.*, 2025). Prior studies indicate that robots can effectively disassemble smartphone modules and unscrew components, which notably increases refurbishment efficiency and enhances worker safety by removing tedious or hazardous tasks, thereby illustrating the principles of human-centric automation (Liu *et al.*, 2026; Lu *et al.*, 2023). Modular refurbishment involves designing products to allow for the replacement of specific modules, such as battery packs or motors, thereby facilitating refurbishment as a process of module replacement rather than complete reconstruction (Tsanakas *et al.*, 2020). The training of technicians in remanufacturing, distinct from line manufacturing, is crucial. Remanufacturing typically involves a more craft-oriented and variable process, necessitating skill in assessing the potential for reuse of components. I5.0 highlights the importance of worker skills, as skilled technicians, aided by digital tools like augmented reality instructions and AI recommendations, can make informed decisions about the reconditioning of parts. This category demonstrates collaboration, as some companies partner with third-party remanufacturers or foster new business ecosystems to oversee their products at the end of their lifecycle. Caterpillar maintains an in-house remanufacturing division, whereas some electronics firms rely on external refurbishing services. Certification programs can be implemented in both scenarios to strengthen quality assurance for refurbished products and support confidence, helping to address perception-related barriers (Regueiro *et al.*, 2024).

3.2.7 Waste-to-Resource Recycling (WRR)

This category includes essential practices that involve the creation of internal recycling loops, which facilitate the direct recycling of scrap and off-cuts within the factory, as well as external recycling initiatives aimed at the collection of products for material recycling. The automobile industry serves as a notable case: end-of-life vehicles undergo dismantling for parts reuse, followed by shredding and material separation to recover metals and other materials. Firms such as BMW are currently designing vehicles with an emphasis on high recycling rates and are investing in recycling technologies that facilitate the recovery of high-value materials (BMW, 2023). I5.0 technology includes advanced sorting methods, such as AI-driven recognition for differentiating various types of plastics, and the use of robotics in recycling facilities to remove contaminants from waste streams, a practice already implemented in some contemporary recycling plants. Chemical recycling of plastics or fibers represents a technological advancement capable of processing mixed or contaminated materials by decomposing them into their fundamental molecular components (R8). Although energy-intensive, when powered by renewable sources and reducing petroleum extraction, it can aid in achieving net-zero emissions. Energy recovery

practices, such as utilizing waste biomass for heat generation in factories or incinerating non-recyclable waste with energy capture, are positioned at the bottom of the hierarchy (Chen, 2018). However, it contributes to net-zero goals if they effectively displace fossil fuels. Carbon capture in waste-to-energy plants integrates net-zero objectives by capturing CO₂ while recovering energy, a scenario currently under exploration by various companies. Recycling presents human-centric challenges, particularly concerning labor conditions in manual processes (Li and Duan, 2025). I5.0 aims to tackle these challenges by implementing automation and improving ergonomics. Moreover, enhancing community engagement through the encouragement of consumer involvement in sorting and returning products requires public awareness, which is fundamentally linked to effective communication.

3.2.8 *Transparent Sustainability Communication (TSC)*

This category signifies an informational layer rather than a physical stage. In this context, practices include eco-labeling products with information about carbon footprint, recycled content, and related attributes. This approach promotes trust and enables informed consumer choices, potentially impacting R0-R2 if consumers choose to reduce consumption or select more sustainable alternatives. Moreover, another approach entails providing recycling instructions via applications or product labeling to improve collection quality, thus enhancing the effectiveness of R8. In internal and B2B contexts, the dissemination of data such as product composition through digital product passports enables recyclers to understand the appropriate treatment of a product (Kemmer *et al.*, 2025). I5.0 technology, specifically blockchain, can guarantee that data regarding product origin and history remains tamper-proof (Fraga-Lamas *et al.*, 2024). This capability is essential for validating claims related to recycled content, thereby preventing greenwashing. QR codes and mobile applications offer an efficient method for communicating detailed information without overcrowding the product with labels. Scanning a code on furniture can provide information regarding the materials used, as well as instructions for disassembly to facilitate moving or recycling, similar to IKEA's approach of offering disassembly guidelines to promote reuse and recycling (Mahalakshmi *et al.*, 2024). The human-centric approach involves customizing communication to meet stakeholder requirements, such as equipping workers with dashboards that consolidate production and sustainability metrics or offering suppliers comparative feedback on their sustainability performance to foster a competitive environment for improvement. Transparency and data flow function as essential facilitators within the system, enabling each participant to contribute effectively to the circular net-zero framework.

3.3 Support System Alignment and Practices

In transitioning to the support categories presented in table 4, we highlight the role these play in facilitating the value chain practices.

3.3.1 *Strategic & Organizational Net-Zero Vision (SONZ)*

This support category involves integrating CE principles into the overarching strategy, culture, and decision-making processes of companies. It encompasses leadership commitment, the establishment of goals for circular performance, and the alignment of the company's vision and values with resource circularity. Implementing ISO 14001, which pertains to environmental

management systems (EMS), or the new ISO 59020 series, focused on CE standards, establishes systematic processes in governance. I5.0 broadens the scope of corporate missions, highlighting the importance of societal well-being through the development of sustainable products (Ghobakhloo *et al.*, 2025). This indicates a shift from exclusive economic goals to an emphasis on the triple-bottom-line framework. A net-zero circular strategy significantly influences all product development decisions, acting as a sustainability gateway within the stage-gate process, which requires that any new product comply with defined circular design criteria. Top management commitment is often recognized in the literature as a critical success factor for I4.0 transformations and sustainability initiatives, thereby increasing its significance when these two domains are combined (Nirmal *et al.*, 2025).

3.3.2 Green Innovation & Technology Development (GITD)

This support category of the value chain promotes investing in new technologies, specifically digital technologies, aimed at enhancing sustainability and fostering innovation in product development. Firms establish specialized teams focused on circular innovation or collaborate with startups through incubators. A manufacturer invests in a startup that specializes in bio-based materials as a possible alternative to current plastics, thus contributing to R0 if it eliminates harmful substances, or R8 if it improves recyclability or composability. (Hwang *et al.*, 2025) Organizing hackathons centered on circular solutions promotes innovative thinking among employees, demonstrating a creative and human-centered approach. Several large size firms have set goals to generate revenue from sustainable products. Their research and development portfolios include projects aimed at low-carbon or circular products. The integration of technology in I5.0 indicates that research and development utilize digital tools extensively, including simulations to assess the effects of recycled materials on product quality and AI in material science to discover alternatives for rare earth metals, thus enhancing resource resilience (Ivanov, 2023). The outcome may include the patenting of a novel recycling process, or the creation of a new product line offered exclusively via a subscription model, thus aligning the business strategy with sustainable design principles. The innovation support category guarantees the company's future viability by investigating fields such as industrial biotechnology, which encompasses the utilization of microbes for metal recovery from electronic waste, a method known as bio-mining (Schiederig *et al.*, 2012). These investments, despite their inherent risks, possess the potential to create transformative capabilities that enable the realization of net-zero circularity.

3.3.3 Regenerative Business Models (RBM)

This category pertains to the implementation process that results in significant transformation. A notable trend in modern practice is the shift from equipment sales to output-based service models. Philips provides lighting as a service for buildings, maintaining ownership of bulbs and fixtures. They facilitate the retrieval of these components and offer essential upgrades, billing customers according to the lumens of light utilized. This method can support improved energy efficiency through the use of high-quality equipment that reduces energy consumption, while also facilitating more structured end-of-life management under contractual arrangement (Howard *et al.*, 2019). Similarly, another model is the product buy-back or trade-in program; numerous firms currently provide customers with discounts for trading in old products when acquiring new ones. This method can facilitate the acquisition of suitable cores for refurbishment

(R5) or repurpose (R7). Automotive manufacturers have utilized trade-in practices for decades; however, smartphone and electronics companies have recently implemented analogous strategies, often refurbishing and reselling products in secondary markets, as demonstrated by Apple's certified refurbished program (Apple Inc., 2013). The alignment with I5.0 requires the implementation of digital platforms for the monitoring of usage, including billing, and for the management of returns. Moreover, these models emphasize the accurate satisfaction of customer requirements via personalization and service orientation, consistent with the goals of I5.0 to tailor production and reduce excess. By selling outcomes, companies reduce the risk of overproduction; they manufacture in alignment with contracted service levels instead of overwhelming the market. Performance contracting represents a developing framework in which a company oversees a client's manufacturing utility systems and is compensated through a portion of the efficiency gains achieved. This arrangement motivates the company to minimize waste and improve equipment durability. Regenerative models entail the design of products for multiple life cycles, facilitating value capture at each stage: revenue from the initial sale, subsequent income from remanufactured sales, and ultimately from the sale of recycled materials (de Souza *et al.*, 2019). Accounting and finance teams must adapt to the complexities associated with multi-cycle revenues and costs, which contrast with one-time sale models. This category requires innovative financing methods, such as leasing arrangements and subscription billing systems.

3.3.4 Synergistic Collaboration Networks (SCN)

This category emphasizes the practical implementation of ecosystem thinking. Implementation can involve participation in industry consortiums, such as CE coalitions or targeted recycling initiatives such as responsible steel consortium or the circular electronics partnership. Collaboration with governments can lead to the establishment of extended producer responsibility laws that advantage proactive companies by fostering a more equitable competitive landscape, such as mandating all firms to participate in recycling initiatives. Ellen MacArthur Foundation's CE100 network, where companies from diverse sectors share best practices related to circular initiatives (Ellen MacArthur Foundation, 2020). Membership in these networks can enhance collaborative efforts. Similarly, at the supply chain level, fostering long-term partnerships rather than transactional relationships enhances the development of collaborative circular solutions. Industrial symbiosis exemplifies a collaborative strategy, as demonstrated by the notable Kalundborg Symbiosis in Denmark (Brieger and Calleja, 2019). This network involves various companies exchanging waste heat, water, and materials, thereby converting waste from one entity into a resource for another, which exemplifies the concept of systemic repurposing. I5.0 technologies enhance communication within networks through a unified data platform for supply chains, enabling comprehensive tracking of materials throughout their lifecycle and allowing each participant to acknowledge the benefits of material return. Collaboration involves the engagement of consumers and local communities, illustrated by partnerships with NGOs focused on consumer awareness initiatives that encourage product returns or reduction in consumption. Initiatives that promote mindful consumption may, paradoxically, be launched by companies that prioritize service rather than focusing solely on increasing sales. Policy engagement entails collaboration with regulators to formulate frameworks that promote circular and net-zero practices, including the advocacy for carbon pricing and the establishment of recycled content standards that support entities already adopting these practices (Yu Ning, 2025).

3.3.5 Sustainability Culture & Skills (SCS)

Transforming organizational culture poses considerable challenges; nonetheless, the advantages become evident when all employees consistently emphasize waste reduction and energy conservation in their daily practices (Davis and Boulet, 2016). Incorporate sustainability into training curricula across all levels, encompassing the onboarding of new employees to communicate the company's sustainability values, enhancing engineers' skills in circular design, and offering courses for factory workers on energy management and lean production with a sustainable emphasis. Numerous organizations establish employee suggestion programs aimed at enhancing environmental practices, with some linking these initiatives to financial incentives or recognition. Gamification strategies improve engagement, as demonstrated by a digital application where teams earn points for implementing waste-reduction competition (Cao *et al.*, 2022). The integration of digital and sustainability skills is crucial; a maintenance technician should develop expertise in operating an AI-driven predictive maintenance system (a digital skill) while also understanding its impact on energy consumption and waste management (a sustainability context). Training programs are progressively incorporating collaborations with universities or online platforms to offer education on topics such as data analytics for sustainability. I5.0 emphasizes a human-centric approach that prioritizes the inclusion of employees' perspectives, advocating for the collaborative development of new work processes with input from operators rather than imposing them unilaterally (Cillo *et al.*, 2022). A significant cultural aspect is the acknowledgment of achievements: sharing internal narratives that emphasize examples such as a team's success in achieving zero waste to landfill in their facility or an engineer's redesign of a component that led to a 50% reduction in carbon emissions, thus positioning these individuals as role models within the organization. This promotes the idea that sustainability is a fundamental responsibility for all individuals, rather than a supplementary task. Moreover, hiring and recruitment strategies may adapt, with the organization potentially favoring candidates who exhibit a sustainability-oriented mindset or pertinent experience. Organizations such as Unilever and Patagonia perform evaluations to assess the compatibility of prospective employees with sustainability values, with the objective of fostering a workforce that inherently comprehends these principles from the beginning (King, 2024).

The support categories constitute the institutional capacity necessary for the effective functioning of value chain practices. Without leadership support and a clear strategic vision, pilot projects are unlikely to scale effectively. Without investment in innovation, the development of new technologies such as tracking systems or improved recycling processes will not occur. Optimizing a linear model without the implementation of new business models may prevent the attainment of complete circularity. Collaboration is essential for closing loops beyond organizational boundaries. In the absence of culture and skills, even the most well-structured plans may fail in execution due to insufficient buy-in or expertise. Each support category corresponds to a specific potential failure point: vision pertains to strategic alignment, innovation relates to technical solutions, business models focus on economic logic, collaboration emphasizes system integration, and culture concerns human execution.

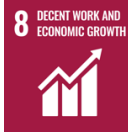
Table 5: Integrated I5.0, circular PLM and net-zero practices alignment with SDGs

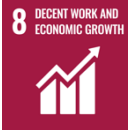
Code	Integrated I5.0 Circular PLM and Net-Zero Practices	R0	R1	R2	R3	R4	R5	R6	R7	R8	R9	SDGs Integration
SPS0101	Buy renewable inputs when its lower life-cycle impacts. Put clear procurement specs and verification in place so “biobased” claims are credible.	✓	✓	✓								13 CLIMATE ACTION 12 RESPONSIBLE CONSUMPTION AND PRODUCTION 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
SPS0102	Use recycled/secondary materials first but verify quality and prove the recycled-content claim. Build specs and supplier documentation so performance and marketing both hold up.			✓						✓		12 RESPONSIBLE CONSUMPTION AND PRODUCTION 13 CLIMATE ACTION
SPS0103	Only use compostable/biodegradable materials when the waste system can truly handle them. Use recognized standards and give clear disposal instructions.	✓								✓		12 RESPONSIBLE CONSUMPTION AND PRODUCTION 13 CLIMATE ACTION
CEDD0201	Make products easy to take apart so they’re easier to repair and easier to recycle. Use reversible joining methods and publish disassembly guidance.					✓	✓	✓				9 INDUSTRY, INNOVATION AND INFRASTRUCTURE 12 RESPONSIBLE CONSUMPTION AND PRODUCTION
CEDD0202	Use standard parts and plug-in modules so repairs and upgrades are fast and affordable. Manage compatibility and spares so the design promise is real.					✓	✓					9 INDUSTRY, INNOVATION AND INFRASTRUCTURE 12 RESPONSIBLE CONSUMPTION AND PRODUCTION
CEDD0203	Build products to last and keep them supported so customers don’t have to replace them early. Durability includes both hardware robustness and ongoing support.	✓		✓								12 RESPONSIBLE CONSUMPTION AND PRODUCTION
CMPE0301	Keep water, heat, and energy circulating inside the plant instead of constantly buying and dumping them. Start with flow mapping, then prioritize practical recovery loops.	✓			✓						✓	7 AFFORDABLE AND CLEAN ENERGY 12 RESPONSIBLE CONSUMPTION AND PRODUCTION 13 CLIMATE ACTION

CMPE0302	Cut waste in factories by fixing root causes like defects and inefficient equipment. Measure energy/material flows and keep improving systematically.			✓								7 AFFORDABLE AND CLEAN ENERGY 12 RESPONSIBLE CONSUMPTION AND PRODUCTION 13 CLIMATE ACTION
CMPE0303	Use less water by reusing suitable water and matching water quality to the job. Track water flows, then optimize and reuse the biggest uses first.				✓							12 RESPONSIBLE CONSUMPTION AND PRODUCTION 13 CLIMATE ACTION
CMPE0304	Make what customers actually need, when they need it, and use digital tools to see problems early. This reduces overproduction and the waste that follows.	✓										8 DECENT WORK AND ECONOMIC GROWTH 12 RESPONSIBLE CONSUMPTION AND PRODUCTION
CMPE0305	Use lean tools to eliminate everyday waste waiting, defects, scrap, and unnecessary steps. Add environmental metrics so improvements reduce both cost and footprint.			✓								9 INDUSTRY, INNOVATION AND INFRASTRUCTURE 12 RESPONSIBLE CONSUMPTION AND PRODUCTION 13 CLIMATE ACTION
GDRS0401	Move freight vehicles to electric (and where appropriate hydrogen), starting with routes that fit. Plan charging/fueling and clean energy supply so the change truly cuts emissions.			✓						✓		7 AFFORDABLE AND CLEAN ENERGY 11 SUSTAINABLE CITIES AND COMMUNITIES 13 CLIMATE ACTION
GDRS0402	Drive fewer miles and burn less fuel using better routing, fuller loads, and driver training. Track results with clear KPIs and keep coaching systematically.			✓								8 DECENT WORK AND ECONOMIC GROWTH 11 SUSTAINABLE CITIES AND COMMUNITIES 12 RESPONSIBLE CONSUMPTION AND PRODUCTION
GDRS0403	Set up practical return channels so used items come back for repair, resale, or recycling. Make returns easy, then sort and route items to the best recovery option.				✓					✓		8 DECENT WORK AND ECONOMIC GROWTH 12 RESPONSIBLE CONSUMPTION AND PRODUCTION

[illegible]

SSE1301	Train people to use sustainability and I5.0 tools in daily work. Skills should be practical data, improvement methods, and safe technology use.	✓											4 QUALITY EDUCATION 	8 DECENT WORK AND ECONOMIC GROWTH 	
SSE1302	Ask employees for circular solutions and fund the best ideas to pilot. Make it real by measuring results and scaling what works.	✓											4 QUALITY EDUCATION 	8 DECENT WORK AND ECONOMIC GROWTH 	
SSE1303	Put sustainability metrics on the same scorecards as cost and quality. Use a few measurable KPIs, track them consistently, and reward verified improvement.	✓											9 INDUSTRY, INNOVATION AND INFRASTRUCTURE 	17 PARTNERSHIPS FOR THE GOALS 	
RES1401	Power operations with renewables, starting by reducing energy demand through efficiency. Plan procurement and reliability so renewable sourcing is credible and stable.									✓			7 AFFORDABLE AND CLEAN ENERGY 	13 CLIMATE ACTION 	
RES1402	Add rooftop solar or on-site wind where it fits your site and load profile. Measure output and integrate it with energy management so the system performs over time.									✓			7 AFFORDABLE AND CLEAN ENERGY 	13 CLIMATE ACTION 	
WRR1501	AI-enabled alloy/polymer loop-closing cell: deploy AI sensing optical/XRF/LIBS, and digital certification to sort in-plant scrap and returned parts by grade, then reintegrate verified recycle into production with embedded carbon factors for net-zero reporting.			✓						✓			9 INDUSTRY, INNOVATION AND INFRASTRUCTURE 	12 RESPONSIBLE CONSUMPTION AND PRODUCTION 	13 CLIMATE ACTION 
WRR1502	Digital industrial symbiosis marketplace: use waste passports and AI matching to route by-products to nearby firms as substitute inputs (with traceability and avoided-emissions accounting), enabling			✓					✓				12 RESPONSIBLE CONSUMPTION AND PRODUCTION 	13 CLIMATE ACTION 	17 PARTNERSHIPS FOR THE GOALS 

	repurposing and reducing virgin-material demand across the ecosystem.											
WRR1503	Real-time monitored energy recovery for residuals: apply pre-sorting, high-efficiency recovery (CHP/WtE), digital twins, and continuous emissions monitoring so non-recyclables provide verified energy that displaces fossil use while maintaining strict emissions controls.			✓						✓		  
SCN1601	Cross-industry symbiosis exchange with carbon traceability: establish a secure collaboration platform that shares material “waste passports” and CO ₂ /by-product specifications across firms, enabling verified substitution of secondary inputs such as capturing and routing CO ₂ streams or mineral by-products while tracking avoided emissions and transport impacts for net-zero reporting.			✓					✓			  
SCN1602	Co-invested regional recovery infrastructure: form a public-private consortium with suppliers, recyclers, local authorities, and peer firms to finance and operate shared take-back, sorting, and advanced recycling capacity for hard-to-recover materials, supported by shared data standards and auditable carbon accounting across participants.			✓						✓		  
SCN1603	pen repair/remanufacture knowledge network: create a governed repository of repair instructions, component design data, and standardized diagnostic interfaces aligned with product passport data standards to enable qualified third			✓		✓	✓	✓				  

	parties to repair, refurbish, and remanufacture products safely, while monitoring quality, worker well-being, and lifecycle emissions benefits across the network.													
RBM0801	Identify hazardous chemicals and replace them with safer options while still meeting performance needs. Document and audit substitutions so improvements stick.	✓												
RBM0802	Use repair and recycling growth as a chance to create inclusive jobs. Provide training and support fair working conditions, and career-advancement pathways.	✓												
RBM0803	Make every repair/recycling/reman activity safe and fair especially when contractors are involved. Use structured safety and labor standards and audit compliance.	✓												
RBM0804	Help smaller suppliers and local firms adopt circular and digital practices through training, tools, and shared infrastructure. This strengthens the whole ecosystem's resilience.	✓												

4. Discussion

The integration of I5.0 principles with circular PLM within a net-zero manufacturing framework produces synergistic advantages that are challenging to achieve through standalone efforts. The resulting configuration enhances sustainable performance and supports competitive advantage by aligning human-centric, resilient, and technology-enabled operations with circular value creation throughout the lifecycle. This section analyzes the principal synergies identified within the framework, explains how the integrated perspective addresses ongoing gaps in the literature, and delineates the broader implications for manufacturing strategy and theory.

A significant tension regarding digitalization. Digital technologies enhance visibility, prediction, traceability, and coordination; however, their mere existence does not guaranty circular or low-carbon performance. Shuang et al. (2026) identified that numerous digital applications in sustainable product management are largely incremental. In contrast, Sarkar et al. (2024) differentiate between the technology-driven focus of I4.0 and the value-driven focus linked to I5.0. Collectively, these findings suggest that the analytical focus should shift from the adoption of a technology to its governance, the lifecycle objectives it serves, and the sustainability outcomes against which it is assessed. The current framework addresses this distinction by considering technology as an enabling layer rather than a sustainability outcome in its own right.

A key synergy relates to resource efficiency, wherein circular strategies such as reuse, repair, refurbish, remanufacture, and recycle are enhanced by advanced digital capabilities. Circular practices aim to separate value creation from the use of virgin resources; however, I5.0 technologies enhance these practices by increasing precision, timeliness, and scalability (Rejeb *et al.*, 2025). Predictive maintenance that integrates IoT sensing with AI-driven diagnostics can forecast equipment deterioration and initiate intervention prior to failure events. This minimizes unplanned downtime and mitigates significant damage that could otherwise decrease asset lifespan, thus strengthen repair and refurbishment strategies while restricting premature replacement. Digitally supported maintenance regimes can mitigate under-maintenance, which results in breakdowns, and over-maintenance, which leads to unnecessary part substitution (Kans and Campos, 2024). This approach effectively reduces material use, energy consumption, and lifecycle costs, thereby achieving sustainability benefits that are economically advantageous.

A secondary synergy emerges from the closure of information loops, facilitating a more dependable closure of material loops. Circular implementation has traditionally been limited by insufficient visibility into the products in use, especially concerning component condition, material composition, and the timing or feasibility of recovery. The concepts of connectivity and the digital thread in I5.0 can mitigate informational asymmetries by integrating in-field data with lifecycle decision-making processes (Fazlollahtabar, 2025). In mobility applications, embedded telematics can facilitate proactive collection or service strategies, such as retrieving traction batteries when health indicators exceed established thresholds. This capability supports remanufacturing, redeployment, or second-life applications, including stationary storage. Digital product passports facilitate end-of-life processing by enabling recyclers to quickly identify alloys, polymers, and hazardous constituents, thereby enhancing sorting quality and recovery yields. Improved data continuity minimizes leakage in circular systems, as components are less frequently discarded due

to unrecognized residual value. Additionally, the prevalence of material downcycling decreases, as sorting decisions are based on informed assessments rather than approximations.

Our proposed integrated framework model suggests a potential advantage in considering operational-energy and material-related emissions within the same lifecycle architecture. Decarbonization initiatives typically emphasize enhancing operational energy efficiency and acquiring renewable energy, concentrating primarily on scope 1 and 2 emissions. CE strategies often focus on embodied emissions and upstream impacts related to materials and product lifecycles (scope 3). The simultaneous implementation of digital process optimization and circular material substitution results in more significant net reductions compared to pursuing either strategy in isolation. Data-driven optimization can reduce energy intensity through improved scheduling, parameter adjustments, and waste minimization (Filipe *et al.*, 2019). Furthermore, enhancing recycled content and adopting life extension strategies can diminish the embodied carbon associated with extraction and primary production. Digital efficiencies often result in prompt cost reductions, which can facilitate the financing of capital-intensive circular initiatives, including reverse logistics, remanufacturing capacity, or advanced recycling (Bag *et al.*, 2021). These relationships are supported both theoretically and empirically in individual studies; however, this review does not provide an estimate of a common effect size, nor does it establish that the integrated configuration will consistently outperform alternative approaches in all manufacturing contexts. Our integrated approach may strengthen the business justification for sustainability, illustrating that environmental and operational performance can be enhanced concurrently, rather than viewed as a necessary compromise.

In addition to emissions and resource performance, integration improves resilience within the manufacturing ecosystem. Circular supply configurations, including localized recovery, diversified input sourcing, and closed-loop material flows, mitigate risks associated with disruptions in virgin supply chains, such as geopolitical events, price fluctuations, and regulatory limitations on essential materials. I5.0 capabilities enhance flexibility through the facilitation of rapid reconfiguration of production systems and the promotion of adaptive decision-making in response to emerging constraints (Wang *et al.*, 2025). Smart factories possess the capability to modify process routes, replace inputs, or redesign components to enhance modularity and repairability, facilitated by digital engineering and real-time performance monitoring. Simultaneously, the human-centric focus of I5.0 gains operational significance: resilience relies not solely on automation and data, but also on a workforce that is trained, empowered, and capable of reconfiguring processes in response to changing conditions (He and Chand, 2024). Cross-trained employees in digital tools and circular objectives exhibit improved ability to sustain continuity amid disruptions and promote rapid learning during transformations.

Integration enhances innovation and differentiation by facilitating the development of new offerings and business models. Circularity can be enhanced via service-based configurations, such as product-as-a-service, which depend on ongoing data exchange, asset monitoring, and lifecycle stewardship capabilities that are closely aligned with I5.0 (Kaur *et al.*, 2026). These models enhance customer relationships, improve retention, and create opportunities for value-added services, such as maintenance, upgrades, and take-back incentives, which are difficult for competitors to replicate without similar lifecycle and data infrastructures. Innovation emerges from interdisciplinary collaboration; reveals practical opportunities for reuse and identifies areas

of waste, while analytics teams uncover optimization patterns that concurrently minimize cycle time and emissions (Ortega Alvarez *et al.*, 2025). The integrated approach serves as a framework for distributed innovation, positioning sustainability objectives as a catalyst for process and product development instead of a constraint.

The proposed framework in this study suggests improvements in stakeholder outcomes, such as customer trust, regulatory preparedness, and brand equity. With the increasing sustainability expectations in both consumer and industrial markets, companies that can effectively showcase their advancements toward circularity and net-zero objectives may gain access to premium market segments, mitigate reputational risks, and enhance their legitimacy with investors and regulatory bodies. Take-back programs, transparent lifecycle disclosures, and verifiable carbon metrics can enhance differentiation, especially when underpinned by digital traceability and auditable data. Furthermore, alignment with emerging policy directions, such as durability requirements, recycled-content targets, and carbon-related trade measures, mitigates compliance risk and enhances strategic continuity. The interplay of these dynamics suggests that environmental sustainability, operational performance, risk management, and innovation can mutually reinforce one another, thereby challenging the traditional view that sustainability primarily represents a cost burden.

4.1 Research Gaps and Theoretical Contributions

This study addresses a persistent fragmentation in the literature by integrating CE strategies, net-zero objectives, the manufacturing ecosystem perspective, and I4.0/5.0 capabilities within a single coherent model. Prior work often treats these domains as parallel streams rather than interdependent components of a unified transformation. The proposed framework contributes theoretically to two main ways. First, it advances I5.0 from a broad policy aspiration toward an implementable operational logic by linking human-centricity, sustainability, and resilience to specific lifecycle practices and enabling technologies. Second, it extends CE theory by embedding circularity within digital transformation and decarbonization imperatives, thereby updating CE scholarship to reflect the contemporary realities of data-intensive manufacturing and escalating climate constraints.

The study presents a framework that builds upon previous integration efforts rather than supplanting them. Atif (2023) conceptualized a data-sharing platform enabled by I4.0, which facilitates digital capabilities to enhance product-service systems and promote value creation within a CE. The present study emphasizes the significance of digital connectivity, lifecycle information, service-oriented value creation, and closed-loop resource utilization as identified in previous research, while broadening the unit and analytical framework from an I4.0-enabled Product-Service System platform to a multilayer manufacturing ecosystem. The extension is developed thru CPLM throughout the entire product lifecycle, incorporating explicit application of R0-R9 strategies, principles of I5.0 such as human-centricity, sustainability, and resilience, as well as dimensions of workforce and stakeholder governance, alongside direct consideration of net-zero management mechanisms. The framework further develops the net-zero circular value-chain architecture proposed by Ikram (2026), maintaining its value-chain and resource-retention principles while enhancing it thru I5.0 governance, a focused CPLM perspective, a digital operating layer, expanded human-centric and ecosystem stakeholder dimensions, and the current

eight-primary/five-support category structure. The contribution is cumulative and integrative; established concepts such as I4.0, PSS, CE, lifecycle, and net-zero are preserved where relevant and reorganized into a comprehensive human-centric manufacturing ecosystem architecture. Some studies approach the problem primarily through the alignment between CE and digital industrial transformation, whereas others focus on sustainable-manufacturing capabilities, strategic enablers, or thematic patterns at the I5.0 and CE interface (Precious, 2025). Across these streams, operational mapping remains less developed than conceptual alignment: the literature identifies technologies and sustainability principles more readily than it specifies how they interact with individual lifecycle stages, R-strategies, organizational responsibilities, and carbon-management practices (Bag et al., 2021; De et al., 2024; Jum'a et al., 2026; Luu et al., 2023).

A notable gap concerns the early stage of I5.0 research, where implementation-oriented frameworks for operations management remain limited. The framework developed in this study specifies how I5.0 principles can be operationalized through technologies mapped to lifecycle stages and circular strategies, while also elevating performance dimensions that are often underrepresented in conventional operations metrics such as worker well-being, reduction of surplus production, and robustness under disruption. The analysis further suggests that I5.0, circularity, and net-zero goals are not independent trajectories; rather, they co-evolve. Human-centric arrangements can stimulate sustainability innovation through workforce involvement, while advanced technologies strengthen circularity by improving traceability and closing loops. This interdependence produces feasible predictions for future empirical research, such as whether plants with greater autonomy and skill intensity attain better circular outcomes due to workers' enhanced ability to recognize opportunities for reuse, repair, and waste reduction.

The ecosystem lens represents an additional theoretical extension. The existing literature on sustainability and green supply chains predominantly focuses on interventions at the firm level. However, achieving net-zero circular transitions often requires coordinated efforts among suppliers, customers, recyclers, and policy stakeholders. The transition is framed as an ecosystem transformation, aligning with systems thinking and contemporary scholarship that emphasizes systemic change, industrial symbiosis, and considerations of a just transition. The proposed framework can function as a structuring tool for subsequent case research, comparative analyses, and survey-based maturity evaluations, facilitating the creation of a net-zero circular readiness model based on technological and organizational aspects.

Furthermore, the adoption of an integrated approach needs that firms evaluate success not only in terms of throughput, cost, and quality, however also through metrics associated with circularity and decarbonization, such as recovery rates, product longevity, recycled-content share, and carbon intensity per functional unit. Financial appraisal practices evolve to incorporate lifecycle costing and carbon accounting, redefining operational excellence as performance within a closed-loop, low-carbon system. This study fills this gap in the literature by integrating approach that introduces complexities that requires careful governance. Advanced digitalization requires significant capital investment and carries an environmental impact; inadequately designed circular strategies may impose short-term productivity challenges. Efficiency gains may lead to rebound effects when decreased costs result in increased output. The identified risks indicate that digital optimization should be explicitly aligned with resource reduction and carbon objectives, utilizing multi-objective decision models instead of solely cost-focused algorithms.

Finally, our proposed model enhances regulatory resilience by developing capabilities that anticipate stricter policy requirements concerning carbon pricing, durability, reparability, recycled content, and extended producer responsibility. From a strategic management viewpoint, the integrated framework serves as a pathway for the development of dynamic capabilities sensing, seizing, and transforming in response to market and regulatory changes driven by sustainability. The integration of I5.0 with circular PLM for net-zero manufacturing establishes a framework that transcends traditional trade-off paradigms. Moreover, it describes a transition where profitability, innovation, and environmental regeneration can be simultaneously achieved, acknowledging that successful implementation relies on intentional design, investment, and organizational transformation.

4.2 Research Implications

This study provides implications for both practitioners and policymakers shaping the enabling environment for sustainable manufacturing. While managerial and policy measures are analytically separated below, the transition is inherently co-produced by firm-level capabilities and supportive institutions.

For manufacturing firms, the framework indicates that implementation should commence with the sustainability issue rather than the technology. Managers should initially identify lifecycle carbon and resource hotspots, assess the technical and economic relevance of various R-strategies, and establish the necessary human and organizational capabilities for their implementation. Digital technologies must be chosen based on specific needs; for instance, IoT facilitates condition visibility, AI aids in prediction and optimization, digital twins enable scenario analysis, and product-passport infrastructure supports lifecycle traceability. This sequence mitigates the risk associated with technology-first investments, wherein firms digitize existing linear processes without significantly enhancing circularity or carbon performance. Performance systems should integrate traditional operational metrics with lifecycle indicators, including energy and carbon intensity, virgin-material displacement, product-life extension, return rates, remanufacturing yield, recovery quality, and workforce-related metrics. The practical implication of I5.0 extends beyond mere automation; it involves a more intentional alignment of technological investment, environmental goals, organizational capacity, and human engagement.

Manufacturing leaders should integrate net-zero and circularity into digital transformation and operational excellence programs, rather than treating them as separate initiatives. Achieving coherence can involve structural integration across functions that have typically been managed in isolation, such as lean operations improvement, digital innovation, and environmental management. A practical mechanism involves establishing a cross-functional governance structure, such as a net-zero and circular transformation office, to support accountability and coordination across technology investments, lifecycle strategies, and performance measurement.

Managers should develop competencies in sustainability analytics, circular design, and inter-firm collaboration. This frequently results in specialized training and the creation of intermediary positions that link data, engineering, and sustainability initiatives such as digital sustainability leadership, circular lifecycle analysis, and ecosystem partnership coordination. The redesign of KPIs and incentives to incorporate circular and net-zero outcomes is equally important.

This includes metrics such as recycled-content ratios, product return rates, remanufacturing throughput, energy mix, carbon intensity, and workforce well-being indicators, ensuring that daily decisions align with lifecycle objectives.

Firms may adjust their portfolio strategy upon identifying products that align with circular business models, such as service-based offerings and take-back programs. These shifts affect pricing, contracting, customer relationship management, and the discourse employed to convey value. Leaders must anticipate the complexities associated with transitions, including reverse logistics, mixed production flows for new and remanufactured components, and evolving inventory dynamics. Effective risk management can be achieved through the implementation of staged pilots and demonstrator lines. I5.0's human-centric orientation necessitates workforce engagement as essential. Transparent communication, reskilling pathways, and structured mechanisms for employee-led improvement enhance implementation quality and reduce resistance to change.

Policymakers can enhance net-zero circular manufacturing through the development of cohesive and integrated policy frameworks that concurrently facilitate digital advancement and the creation of circular value. Incentives tend to be most effective when they provide rewards for dual outcomes, such as digital investments that clearly enhance resource productivity, recovery performance, or lifecycle emissions. Public support is essential for the development of infrastructure, including advanced recycling and remanufacturing capabilities, reverse logistics networks, and interoperable digital standards that enhance lifecycle traceability.

Regulatory measures can strengthen market signals by establishing expectations for durability, repairability, recycled content, and disclosure, while technical assistance can help smaller firms participate more effectively in the transition. Workforce policy is crucial; vocational and higher education curricula must align with the hybrid skill profile necessary for digital-circular operations, underpinned by certifications and apprenticeship models that professionalize new roles in remanufacturing and circular analytics. Market-based instruments, such as carbon pricing and extended producer responsibility, can effectively realign incentives towards low-carbon and circular configurations. Additionally, public procurement can stimulate demand for products that adhere to established circularity and footprint standards.

Furthermore, ensuring policy coherence among climate strategy, industrial modernization, digital strategy, and education policy is crucial. Coordinated action can diminish uncertainty, decrease transition costs, and encourage investment, facilitating the shift of firms from incremental improvements to comprehensive, ecosystem-level transformation aimed at achieving net-zero circular manufacturing.

5. Conclusion

This study presented a theoretical model explaining how manufacturing firms can jointly operationalize I5.0, r CPLM, and net-zero carbon objectives through a single, value-chain anchored framework. By integrating a bibliometric analysis with a systematic literature review the proposed model builds on 50 integrated I5.0, circular PLM, and net-zero practices, structured across eight value-chain (primary) categories and five support and net-zero categories, so that

renewable energy adoption, carbon measurement and traceability, and emissions-reduction actions are embedded at each stage of the product and process lifecycle. The main contribution of this work is the integration of two domains that are typically treated as separate: (i) CE strategies organized around R0-R9 resource loops, and (ii) corporate carbon management approaches aimed at measurement, mitigation, and net-zero alignment. We systematized how each circular value-chain and PLM practice can be adapted, extended, or reoriented to deliver direct greenhouse-gas reductions, and we mapped the relationships between these adapted practices and the relevant R-strategies. By doing so, the framework clarifies where particular circular interventions are most directly connected to plausible climate-mitigation mechanisms and where carbon-focused measures may reinforce circular performance. Because the review does not apply a common quantitative comparison of emission-reduction effects, the framework does not rank practices according to 'largest' climate. Our proposed model therefore advances sustainable operations theory by showing, in an actionable structure, that resource circularity and carbon neutrality are mutually reinforcing outcomes when pursued through a coordinated, lifecycle-based strategy rather than isolated initiatives.

Manufacturing in the twenty-first century is significantly influenced by two concurrent transitions: the imperative for deep decarbonization and the swift pace of technological advancement. This study presents a model that conceptualizes the transition as a comprehensive redesign of manufacturing ecosystems at the system level, rather than merely an incremental series of technical enhancements. Circularity is framed as both a materials strategy and a carbon strategy, as it can significantly reduce lifecycle emissions by decreasing the demand for virgin materials and prolonging the lifespan of products and assets. Simultaneously, it considers I5.0 capabilities connectivity, data-driven decision support, and human-centered work design as the essential conditions that facilitate large-scale closed-loop coordination.

The proposed model suggests that digital transformation and sustainability initiatives ought to be managed as a unified agenda. The proposed structure illustrates that coordinated interventions across design, sourcing, production, distribution, use, and recovery can yield both environmental and competitive advantages, rather than viewing I4.0/I5.0 solely as a productivity pathway and circularity/net-zero merely as a compliance pathway. From this perspective, lifecycle stages serve as critical decision points for firms to enhance value retention via repair, refurbishment, remanufacturing, repurposing, and recycling, while also minimizing emissions through low-carbon energy, process optimization, and transparent carbon tracking. I5.0's focus on human-centric principles underscores that the transition extends beyond mere automation; it necessitates skills development, active worker involvement in enhancements, and collaboration among stakeholders to maintain enduring change.

Furthermore, this study offers a structured approach for evaluating readiness and prioritizing interventions from a managerial viewpoint. This framework supports decision-making related to the incorporation of circular-by-design principles, the creation of data continuity for lifecycle traceability, the synchronization of recovery systems with operational planning, and the

expansion of business model innovations, such as product-service systems and take-back arrangements. The model suggests that attaining net-zero progress is unlikely through isolated initiatives; it requires coordinated transformations across product development, supply chain governance, manufacturing processes, and customer engagement. Organizations that emphasize early investment in integrated traceability, recovery pathways, and emissions management are likely to improve operational resilience and gain strategic advantage, especially as markets and regulations increasingly support low-carbon and resource-secure production methods. The model highlights the importance of unified policy frameworks that align industrial modernization with climate and circularity objectives. Standards that promote traceability, incentives for investments that integrate digital capabilities with circular outcomes, and infrastructure that supports recovery and renewable integration can improve diffusion and mitigate adoption risk, particularly for smaller firms. The model provides a standardized framework for industry and regulators by outlining the components of change and the interconnections among circular performance, carbon outcomes, and enabling technologies.

5.1 Limitations and Future Research Directions

Despite its comprehensive nature, this study proposed a conceptual model and thus has several limitations. The framework has been constructed through a synthesis of literature and conceptual reasoning, rather than through primary empirical validation. The explanatory adequacy, completeness, and sectoral relevance should be evaluated in various manufacturing contexts, including discrete high-volume production, capital-intensive equipment industries, and process sectors. Future research should utilize case study and action research methodologies to analyze implementation pathways, pinpoint operational bottlenecks such as regulatory and technological limitations, and evaluate quantifiable outcomes. Comparative multi-case research is essential for identifying elements that are generalizable across contexts versus those that necessitate sector-specific adaptations.

The framework's extensive scope, although beneficial for conceptual integration, may pose challenges for practical adoption. This discussion highlights complementarities: however, practical implementation may entail tensions and trade-offs. Extending product life (R3-R6) may pose challenges to revenue models based on new-product sales. Additionally, pervasive sensing and tracking can lead to concerns regarding cybersecurity, data governance, and privacy. Furthermore, the environmental footprint of digital infrastructure, including devices, networks, and data centers, may partially negate sustainability gains if not managed effectively. Subsequent research should advance beyond qualitative claims of synergy to quantify the conditions that lead to net benefits, including the identification of thresholds at which rebound effects occur. System dynamics modeling, lifecycle costing, and integrated lifecycle assessment can elucidate the co-evolution of economic and environmental performance over time within digitally enabled circular systems.

The proposed framework does not explicitly integrate external behavioral and market contingencies. Various circular strategies depend on customer acceptance, including the adoption of remanufactured products, participation in take-back schemes, and the shift towards access-based consumption. Future research should incorporate studies on consumer behavior and perform behavioral experiments to clarify adoption barriers, evaluate the effectiveness of incentive mechanisms, and investigate the impact of trust and perceived quality. Suppliers conduct and power imbalances similarly limit upstream circular procurement. Future research may explore the alignment of incentives and the dynamics of bargaining in circular supply arrangements, utilizing analytical approaches like game theory and contract design.

Finally, the proposed framework in this study should be regarded as a systematic platform for continuous investigation rather than a conclusive directive. Net-zero circular manufacturing represents a dynamic, multi-faceted transformation influenced by technological advancements, policy developments, and societal expectations. Future research should aim to evaluate, analyze, and expand the framework to assess its robustness and practical relevance in guiding manufacturing ecosystems toward credible sustainability transitions.

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